Design System & UX

polycarpic — personal budgeting & wealth management (virtual Single Family Office)

Owner: `ux-designer` agent **Status:** Draft — Plan phase, matured alongside ARCH **Version:** 0.15

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1. Overview & Design Principles

What this product should feel like to use, and the principles that resolve design disputes.

- **Mobile-first when in doubt.** The primary v1 target is a self-hosted web dashboard used on desktop for deep review (statements, lot tables) and on mobile for quick checks (dashboard glance, confirming a draft transaction) — design the dense screens up from a mobile layout, not down from desktop.
- **Accessible by default (WCAG AA).** Non-negotiable per PRD §6. Financial gain/loss and status signals never rely on color alone — see Financial semantics.
- **No decorative loading states.** Skeletons match the real layout; spinners are a last resort (see Component inventory).
- **Every number earns trust.** This is a general ledger, not a budgeting toy — a balance, a delta, or a pending-liability figure must always be traceable to its source (the journal entry, the lot, the reconciliation). The UI never shows a number it can't explain if clicked.
- **Immutability is visible, not just enforced.** A posted transaction reads as posted (badge, disabled edit affordance); a correction reads as a reversal + new entry, never as a silent edit (PRD §5).
- **Diverge from polycarpic's own tooling, deliberately.** The cairn tracker (our internal tool) and the product are not the same thing and shouldn't look like it — this system's inspiration is derived from measured screenshots and live style reads off Monarch Money and Origin — see Inspiration: Monarch & Origin.
- **One brand hue, used honestly.** The neon-green brand color (`#2CFF05`, "the project name relates to fruiting plants") measures 1.36:1 against white, far below AA — kept as the literal `--brand` value everywhere it clears AA (dark surfaces), with a darkened same-hue green for light-mode `--primary`, stated plainly rather than silently swapped for a different hue — see Foundations.

2. Design System

The single source of truth for the visual language. The written spec lives alongside this doc; the machine-readable system lives in the two CSS files that `frontend-lead` consumes directly.

- design-system-spec.md § Inspiration — what was taken from Monarch Money and Origin, and why
- design-system-spec.md § shadcn-svelte reuse map — every component/region mapped to its shadcn-svelte component, block, or a stated "custom" reason
- design-system-spec.md — the written spec (token rationale, financial semantics, component inventory, accessibility)
- tokens.css — design tokens as CSS custom properties (hand-authored palette, not a vendored shadcn preset)
- tokens.html — every token decoded: name, value in both modes, swatch, one-line "used for"
- screen.css — component & screen styles built on the tokens

`docs/DESIGN/` is the in-repo, code-level home that ships with the project and is the contract for implementation.

3. Design Tokens

Color, typography, spacing, radius, and shadow primitives — SvelteKit + shadcn-svelte (PRD §8). Edit values in tokens.css; the swatches below preview the live variables. Full rationale in design-system-spec.md.

> **Token reference** Every token with light & dark hex values, swatches, and "used for" →

3.1 Surfaces & Text



`--background` `--foreground` `--card` `--popover` `--muted` `--muted-foreground`

3.2 Actions & States



`--primary` `--brand (dark surfaces only — see Foundations)` `--brand-2 (same rule — see Secondary accent)` `--secondary` `--accent` `--destructive`

3.3 Financial semantics (new — polycarpic-specific)



`--positive` `--warning` `--destructive (reused for negative/danger)`

See design-system-spec.md § Financial semantics — gain/loss, safe-harbor status tiers, and the "never color alone" rule.

3.4 Borders & Focus



`--border` `--input` `--ring`

3.5 Charts (sequential ramp of --brand-2 — magnitude, not category)

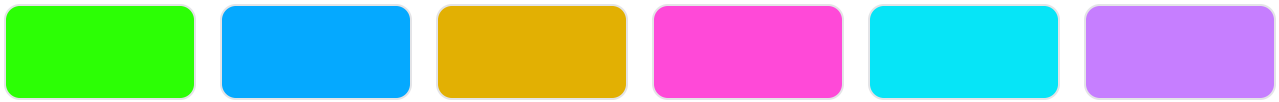




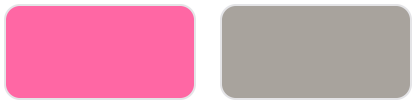
`--chart-1` `--chart-2` `--chart-3` `--chart-4` `--chart-5`

3.6 Categorical (identity, not magnitude)

Seven fixed-order hues, the literal `--brand` and `--brand-2` plus five at their brightness and intensity, and a neutral Other, for treemaps, Pack, Sankey and bars. Colour follows the entity, never its rank. The light-mode 3:1 surface rule is waived for these fills, compensated by a label, a hairline or gap, and dark label ink. Values, label inks and fixed assignments: tokens.html § 7; method and validator results: design-system-spec.md § Categorical palette.

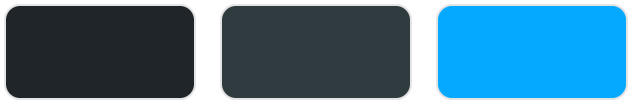


`--cat-1` `--cat-2` `--cat-3` `--cat-4` `--cat-5` `--cat-6`



`--cat-7` `--cat-other`

3.7 Sidebar (inverted, blue-tinted)



`--sidebar` `--sidebar-accent` `--sidebar-ring`

Dark-on-light in light mode, light-on-dark in dark mode — see design-system-spec.md § Sidebar for the collapse-to-rail behavior and three proposed expand/collapse cues.

Radius: *measured `--radius-sm|md|lg|xl` = 6/8/12/20px (cards 12px, buttons 8px — both products' actual app-level values, not either's marketing-only pill).*

4. Components & Screen Styles

The component inventory and the styles that realize it, built on the tokens. Styles live in screen.css; full states-per-component table lives in design-system-spec.md § Component inventory.

Component	shadcn-svelte primitive	polycarpic usage
Net-worth hero	Chart (LayerChart, Area)	Big figure + signed delta on the figure + single-line area chart — Monarch-inspired
Grouped account list	Table / custom	Type-grouped rows with sparkline + freshness — Monarch-inspired
Segmented bar + legend	Custom, tokens-driven	Assets/Liabilities, allocation actual-vs-target — Monarch-inspired
Badge	Badge	Entity/account/transaction status, link-health — see Financial semantics
Transaction table	Table	Account register, exceptions report, lots
Accordion	Accordion	Pending row expansion
Tabs	Tabs	Transactions' stages with count badges; General Ledger's sections; Reports' statements; Tax's jurisdictions
Near-full-screen modal	Dialog	Journal-entry modal (header + legs) — manual entry, Correct entry, Split, Group & Summarize
Entity switcher	Tabs / Sidebar	Switch between Person/Trust/Business/Household

Component	shadcn-svelte primitive	polycarpic usage
Transaction detail	Sheet	Drill into a posted/draft entry, its journal lines, source doc
Empty state	Card + plain text, no skeleton	Dashboard panels not yet shipped (tax at 1.0, allocation at M4) — see styled-screens/dashboard.html
Top row	—, starts right of the sidebar	Not sticky, same background as content: breadcrumb left (expand/collapse lives in the sidebar brand row); entity selector + separator + "more" menu right — AI/search/notifications/gear dropped — see design-system-spec.md § Header
Sidebar	Sidebar (shadcn-svelte sidebar-07 block)	Floating card, gutter + rounded all 4 corners, eight nav items, brand top item carrying the expand/collapse control (`panel-left-close` right-aligned; collapsed, the mark swaps to `panel-left-open` on hover/focus), avatar-popover bottom item opening right — see design-system-spec.md § Sidebar
"More" menu	Dropdown Menu	Top row, right-justified — per-page global actions (Sync, New entry, Export, ...)
Avatar cluster	—, custom	Overlapping initials in the entity selector, max 3 + overflow
Avatar popover	Popover / Dropdown Menu	User menu at the sidebar's bottom, opens to the right — profile, Settings (proposed), Notifications (proposed), preferences, theme, keyboard shortcuts, Log out last
Landing illustration	—, custom	Home — the mountains illustration under the brand app bar, then headline, subhead and feature cards; not a signed-in screen
Switch	Switch	Settings on/off preferences (auto-post matched recurring transactions, reduced motion)

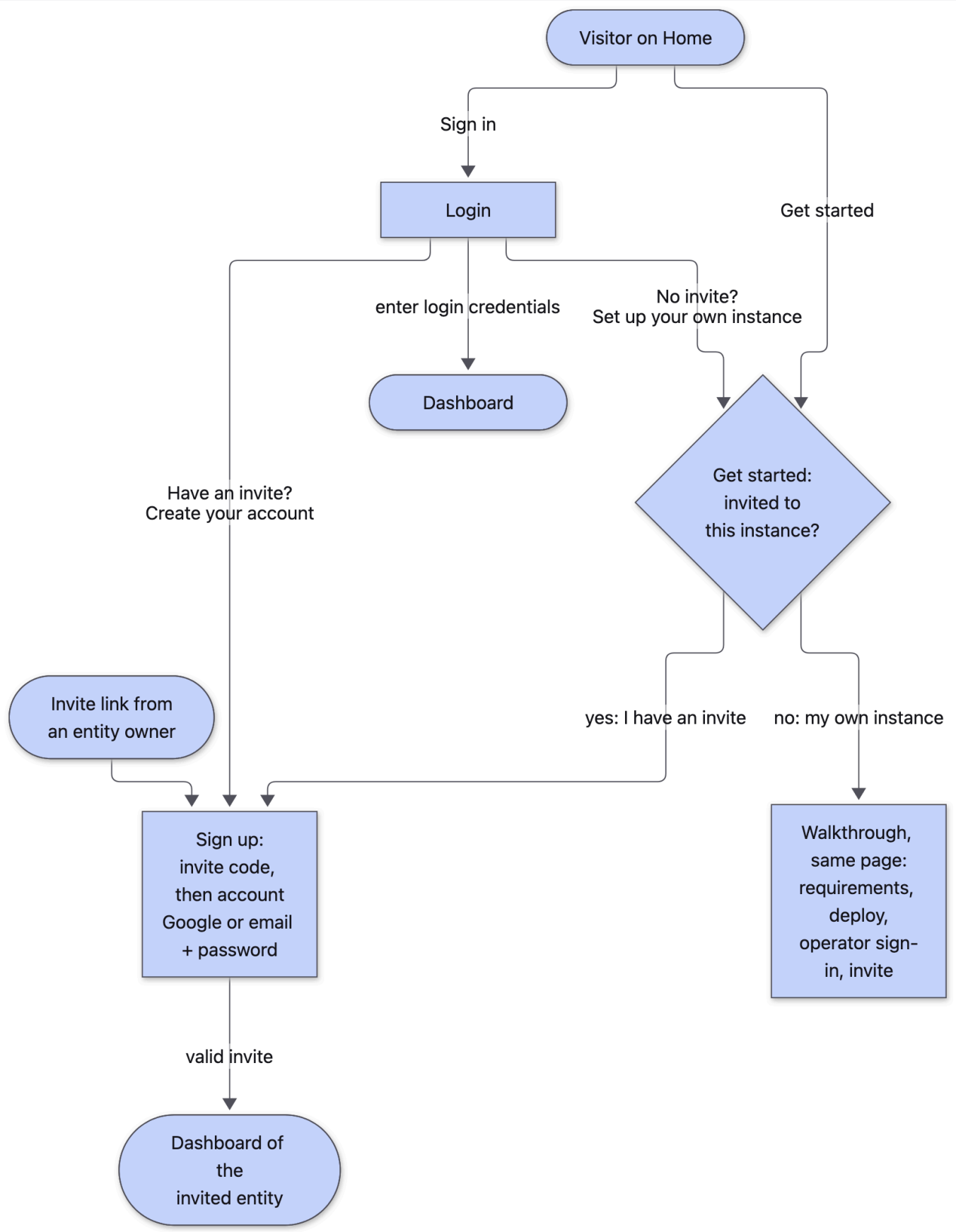
Component	shadcn-svelte primitive	polycarpic usage
Chart-of-accounts tree	—, custom (nested disclosure rows)	General Ledger: group rows with chevrons, account rows link to the register, custodial cross-reference badge
Date range control	Popover + Range Calendar	General Ledger's Journal entries card: presets plus one month grid
Journal entry popover	Popover	Transactions' Posted tab: a Journal ID opens the entry in place, with its legs, correction trail and a link to the General Ledger
Brand app bar	—, custom	Every page without the app shell (Home, Get started, Admin): fixed near-black bar, neon lockup left, the page's trailing items right, a fade below (deep over Home's image, a short shadow over plain pages)
Split screen login	—, custom	Login — full field list, centered form column, always-branded visual pane

5. User Flows

How users move through the product to accomplish the highest-traffic stories (PRD §4). Each flow is embedded inline below, because a browser serves `flows/*.md` as raw Markdown and its fenced `mermaid` blocks would never reach the page's Mermaid loader. The `.md` files stay as the diffable source of truth in flows/.

5.1 Home entry: Sign in, Get started, Sign up

Story (PRD §4.1): *a visitor's first move from Home, on a product with no public sign-up. Source: flows/home-entry-flow.md.*

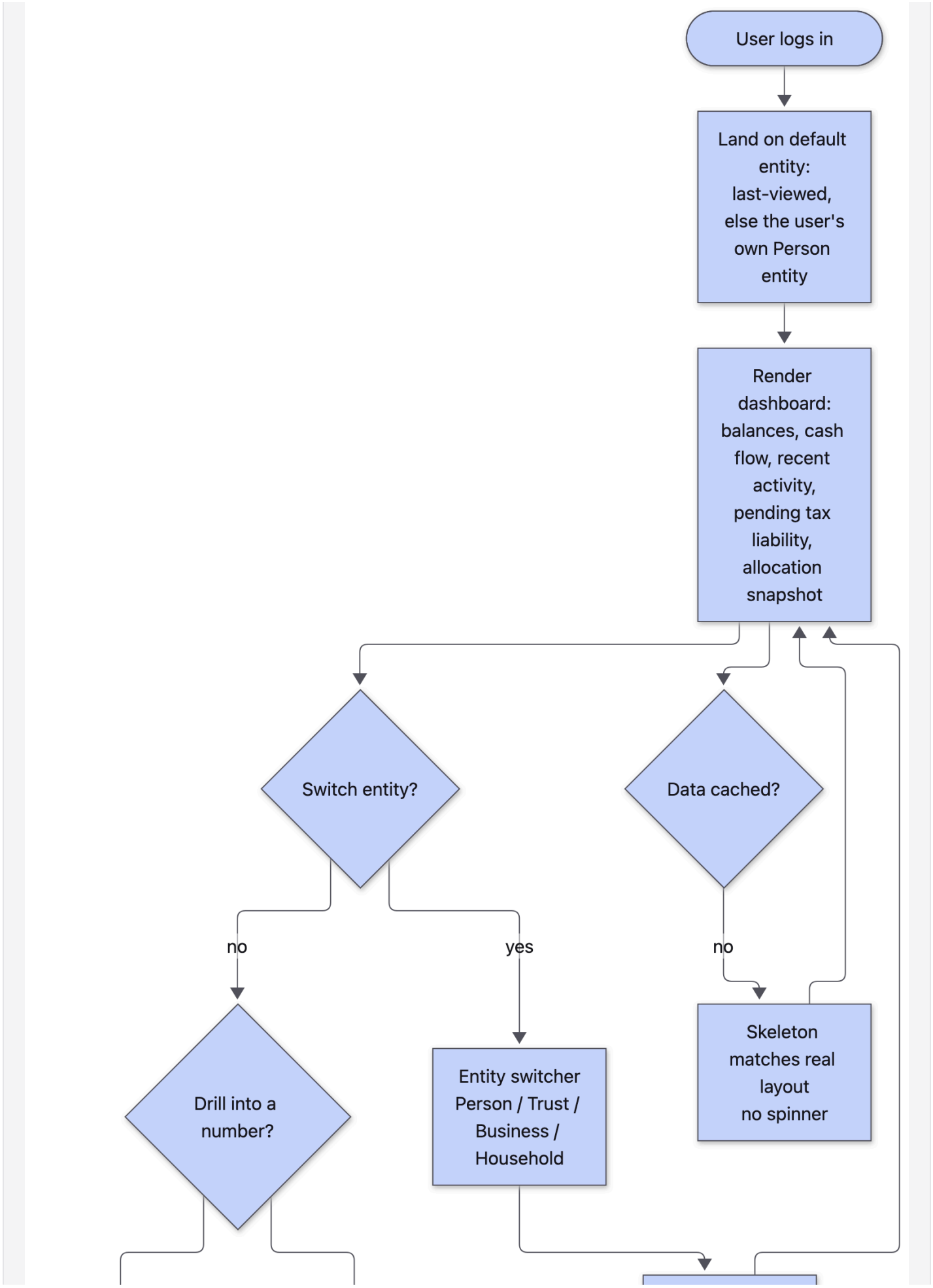


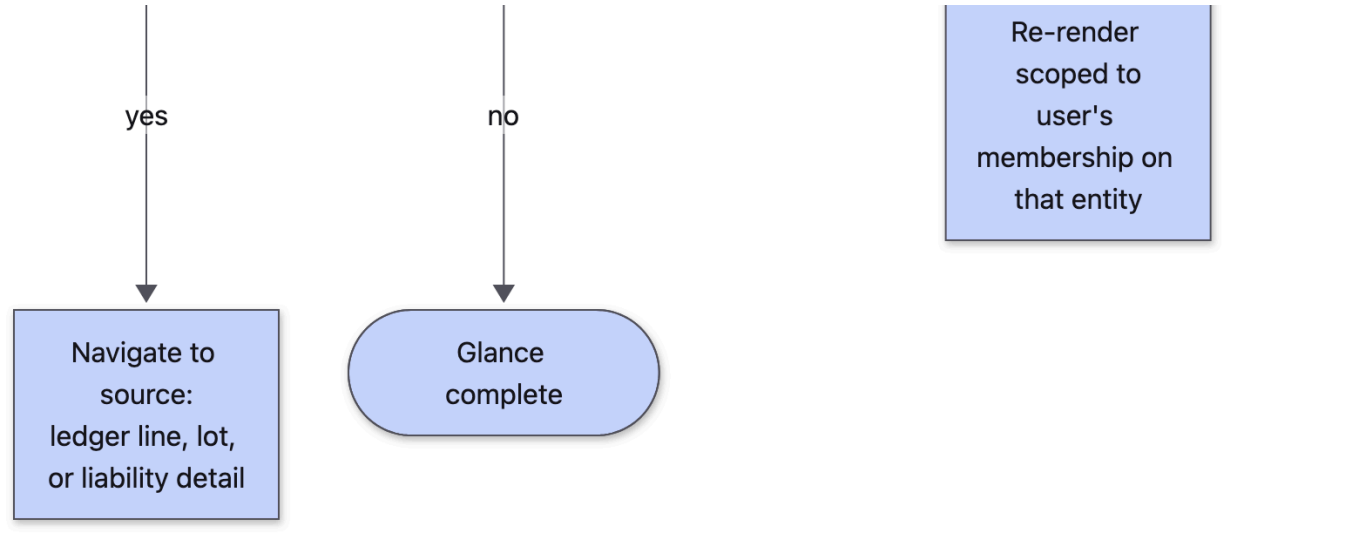
- **No public sign-up.** Every instance is self-hosted and accounts on it come by invitation from an entity owner. The bar's action is therefore "Get started", never "Sign up".

- **Get started is one page, not a modal.** The fork sits on top as two cards ("I have an invite" → Sign up; "I'm setting up my own instance" → the walkthrough directly below), so a self-hosting visitor never leaves the page and the walkthrough has room to grow. *Alternative:* a modal from Home's Get started button holding just the two cards; lighter on Home, but the walkthrough then needs its own page anyway.
- **Login shows the same branch.** Its account line reads "Have an invite? **Create your account**" (to Sign up) and "No invite? **Set up your own instance**" (to Get started).
- **Every sign-in lands on a dashboard.** Account creation also creates the user's own Person entity, so there is no signed-in state without an entity (design-system-spec.md § Entities & membership). Login lands on the last-viewed entity, else that Person entity; the operator's first sign-in after setup lands on the operator's own Person entity.
- **Sign up is invite-gated.** The code arrives pre-filled and validated from an invite link, or is typed. A valid invite names the inviter and the entity. The invite is common to both account paths (Google, or email + password), and creating the account lands on the dashboard of the invited entity.
- An invalid or used code keeps the visitor on Sign up with an inline error under the field; it never sends them to a dead end.

5.2 Dashboard glance

Story (PRD §4.2): *per-entity financial health at a glance. Source: flows/dashboard-glance-flow.md.*

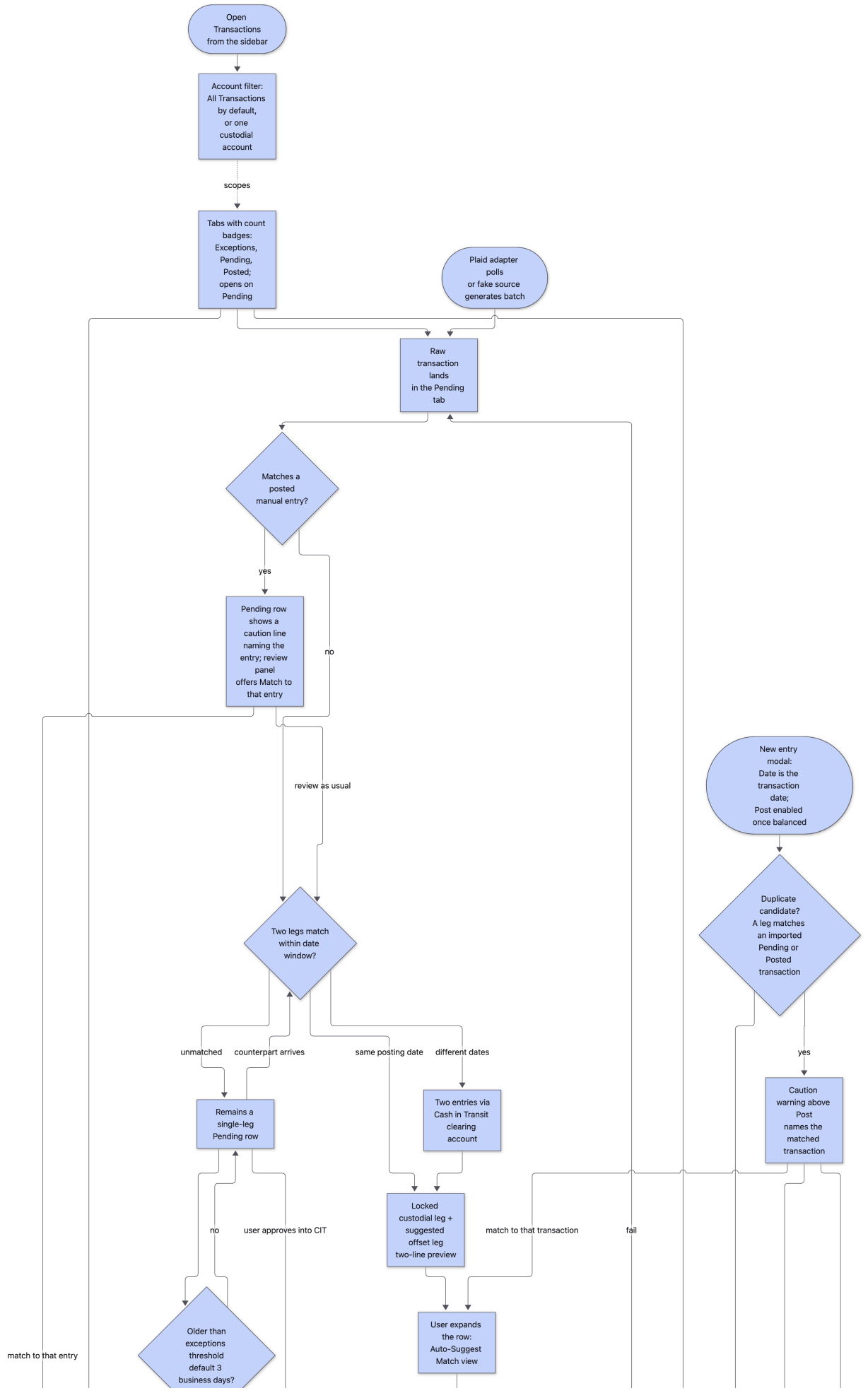

User logs in
Land on default entity: last-viewed, else the user's own Person entity
Render dashboard: balances, cash flow, recent activity, pending tax liability, allocation snapshot
Switch entity?
Data cached?
no
yes
no
Drill into a number?
Entity switcher Person / Trust / Business / Household
Skeleton matches real layout no spinner

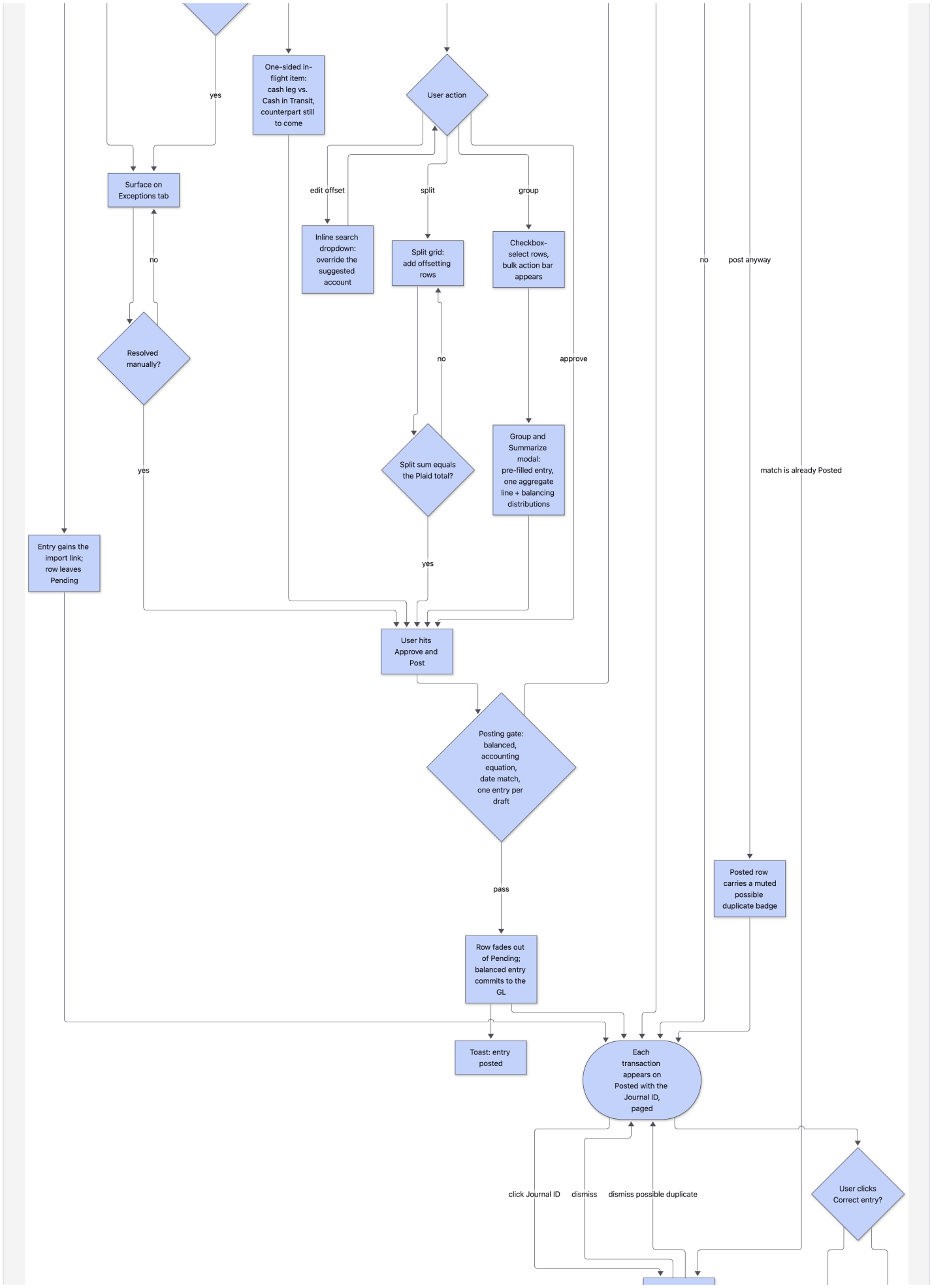


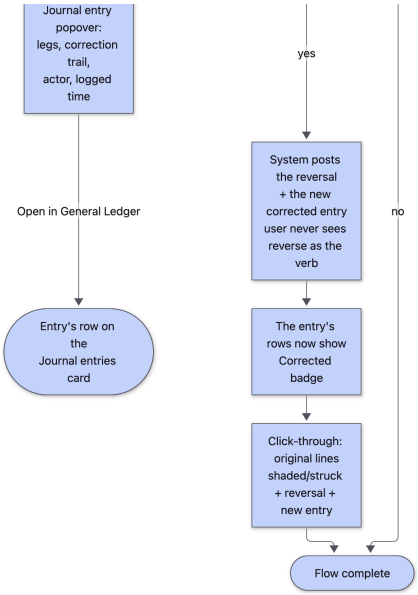
- Every number on this screen is clickable through to its source (Overview principle: "every number earns trust").
- Filtered strictly by membership (owner / manager / viewer) — no transitive access via a parent entity (PRD §5 permission model).
- Viewer-role household members see the household's shared accounts and their own entity, never another member's entity (PRD §4.9).
- **"Skeleton matches real layout, no spinner":** while dashboard data is loading, the screen shows placeholder blocks already shaped like the real cards (same card chrome, same approximate text-line widths) instead of a spinner, so nothing jumps or re-flows when the real numbers arrive; they fade in where the placeholders were. A spinner says "wait"; a layout-shaped skeleton also says what is coming. See design-system-spec.md § Component inventory → Skeleton.

5.3 Transactions — review, match & correct

Story (PRD §4.3): *staged for review → matched / split / grouped → posted → correctable. The review mechanics follow a Plaid bank-feed workflow: every imported transaction is reviewed, matched, split or grouped, then posted. Source: flows/import-confirm-flow.md.*


Open Transactions from the sidebar
Account filter: All Transactions by default, or one custodial account
scopes
Tabs with count badges: Exceptions, Pending, Posted; opens on Pending
Plaid adapter polls or fake source generates batch
Raw transaction lands in the Pending tab
Matches a posted manual entry?
yes
no
Pending row shows a caution line naming the entry; review panel offers Match to that entry
review as usual
match to that entry
Two legs match within date window?
unmatched
counterpart arrives
same posting date
different dates
Remains a single-leg Pending row
Two entries via Cash in Transit clearing account
no
user approves into CIT
Locked custodial leg + suggested offset leg two-line preview
match to that transaction
fail
Older than exceptions threshold default 3 business days?
User expands the row: Auto-Suggest Match view
New entry modal: Date is the transaction date; Post enabled once balanced
Duplicate candidate? A leg matches an imported Pending or Posted transaction
yes
Caution warning above Post names the matched transaction


yes
One-sided in-flight item: cash leg vs. Cash in Transit, counterpart still to come
User action
Surface on Exceptions tab
edit offset
split
group
Inline search dropdown: override the suggested account
Split grid: add offsetting rows
Checkbox-select rows, bulk action bar appears
no
no
post anyway
Resolved manually?
no
approve
Split sum equals the Plaid total?
Group and Summarize modal: pre-filled entry, one aggregate line + balancing distributions
yes
match is already Posted
Entry gains the import link; row leaves Pending
yes
User hits Approve and Post
Posting gate: balanced, accounting equation, date match, one entry per draft
Posted row carries a muted possible duplicate badge
pass
Row fades out of Pending; balanced entry commits to the GL
Toast: entry posted
Each transaction appears on Posted with the Journal ID, paged
click Journal ID
dismiss
dismiss possible duplicate
User clicks Correct entry?



- **Account filter.** A selector at the right end of the tab row scopes Exceptions, Pending and Posted together, count badges included. Opening Transactions from the sidebar always lands on **All Transactions** (every custodial account in the entity); the popover lists each custodial account by the name Custodial Accounts uses, with its provider and open-item counts. The Known delta badge on a Custodial Accounts card deep-links here with the filter preset to that account. On Posted the rule is leg-level and the same for every row: with one account selected, only the rows whose own leg posts a debit or credit to that account show, so the counterpart row of a transfer does not appear. A manual entry with a leg on that account shows as one row for that leg. All Transactions shows everything, including a manual entry with no custodial leg, such as an accrual, which no single account's view lists.
- **Tabs with count badges.** Exceptions, Pending and Posted are one `Tabs` row, left of the account filter, one visible at a time, in urgency order. Each tab carries its count, so every stage's load stays visible while another is open: `secondary`, `destructive-soft` on Exceptions when non-zero. Transactions opens on Pending, the review inbox; a non-zero Exceptions count does not take the default, its badge is red instead. The counts follow the account filter, changing the filter keeps the current tab, and the URL hash carries the tab.
- **Every row names its source.** Each Exceptions, Pending and Posted row carries one muted line under the payee: `landmark` icon + `<account> · <institution> via <provider>` (e.g. "Primary Checking · Bank Y via Plaid"). A manual entry has no bank source, so its line reads `pen-line` icon + "Manual entry". With the filter on one account, a line that only repeats the trigger is omitted; a Manual entry line stays.
- **The locked leg is always the custodial cash side,** debit on inflows, credit on outflows — verified bank reality (the Plaid transaction itself), never editable, in the two-line preview, the Split grid, or the Group modal. **The suggested offset leg** is pre-filled from historical rules + the Plaid category and overridden in place via an inline search dropdown, never a blank field.

- **A leg always names its account by full path,** root to leaf, in the Pending preview, the Split grid, the New entry modal, and the Posted expansion and popover: Expenses › Office Expense. Account controls show and list paths.
- **Split** is one draft → one multi-leg entry, live-validated: submission stays disabled until the rows sum exactly to the raw transaction's absolute total. **Group** is many drafts → one entry (ARCH §3), the same many-to-one shape as a same-date matched transfer, via the shared journal-entry modal, pre-filled with one aggregate line + balancing distributions.
- **Posted is transaction-scoped.** Every posted transaction is a row, mirroring Pending, and carries its entry's Journal ID. An entry built from several transactions appears as several rows with one ID: a group, or a same-date matched transfer (two legs, one entry). The rows sit together, because an entry has one date and same-date rows order by Journal ID. A split stays one row, however many legs its entry has. A transfer whose legs land on different dates posts as two entries through Cash in Transit, so its two rows carry two IDs. With the filter on one account, only that account's row of a transfer shows, whether the transfer is one entry or two; the Journal ID popover still lists both legs. The Posted badge counts rows, so it differs from the entry count on the GL Journal entries card, which is the entry-grained view.
- **A manual entry is a Posted row like any other.** On All Transactions, an entry from the New entry modal is one row with its own Journal ID, however many legs it has, and its Amount is the entry's sum of debits, unsigned, because it has no bank sign. With one account selected it is one row for the leg on that account, signed like a bank row: a debit to a cash account reads +. It cannot exist outside a compliant journal entry, so the same rules apply: the account filter, Correct entry, the Journal ID popover.
- **Row Date is the transaction date.** Date on Posted and on the GL Journal entries card is the date the user entered on the entry, never the time it was logged. The New entry modal's Date field is that date, so a backdated entry sorts by it on both lists, while its Journal ID still orders by posting. The audit trail (actor, logged-at) lives in the Journal ID popover and the GL entry view, not in the list.
- **Potential-duplicate warning.** While a manual entry is open in the New entry modal, each custodial leg is compared with imported transactions that are Pending or Posted. A candidate has the same custodial account, the same amount and direction, and a date within the entity's Duplicate detection window of the entry's Date, either side (Settings, default ±1 day). A candidate shows a caution-tier warning above the actions naming it: "Looks like Sep 12 · Stripe Payout +$1,500.00 on Primary Checking, Pending". The nearest date is named, and a Pending candidate wins a tie. **Match to that transaction instead** discards the manual entry and opens the Pending row's review panel, pre-filled with the modal's offset leg; for a Posted candidate this action reads **Open that entry** and opens its Journal ID popover. **Post anyway** takes the Post button's place and posts the entry: its row carries a muted `outline` `possible`

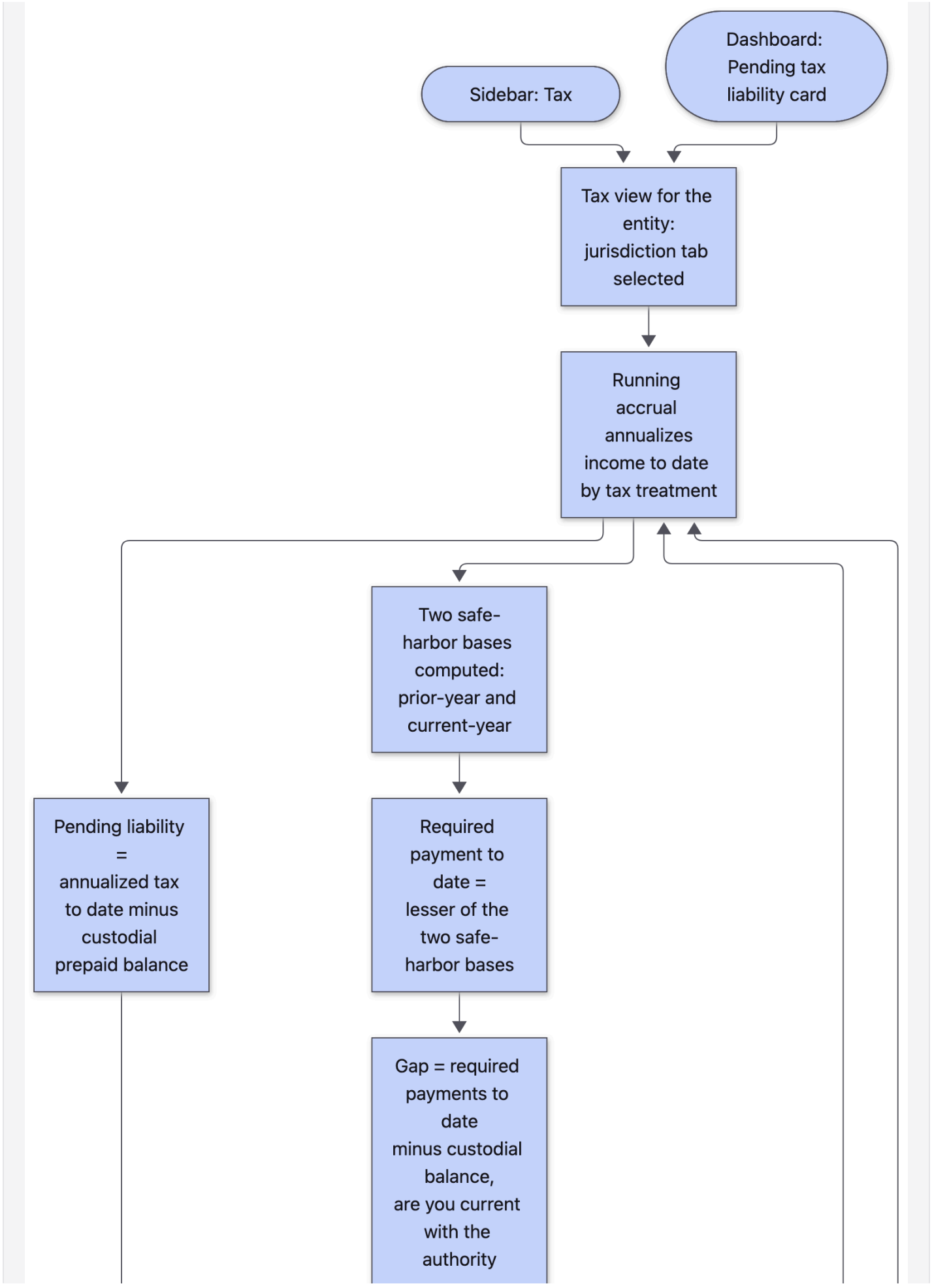
`duplicate` badge under its status until the user dismisses it from the popover or matches the import. Only New entry and imports check, because Correct entry, Split and Group start from a real transaction.

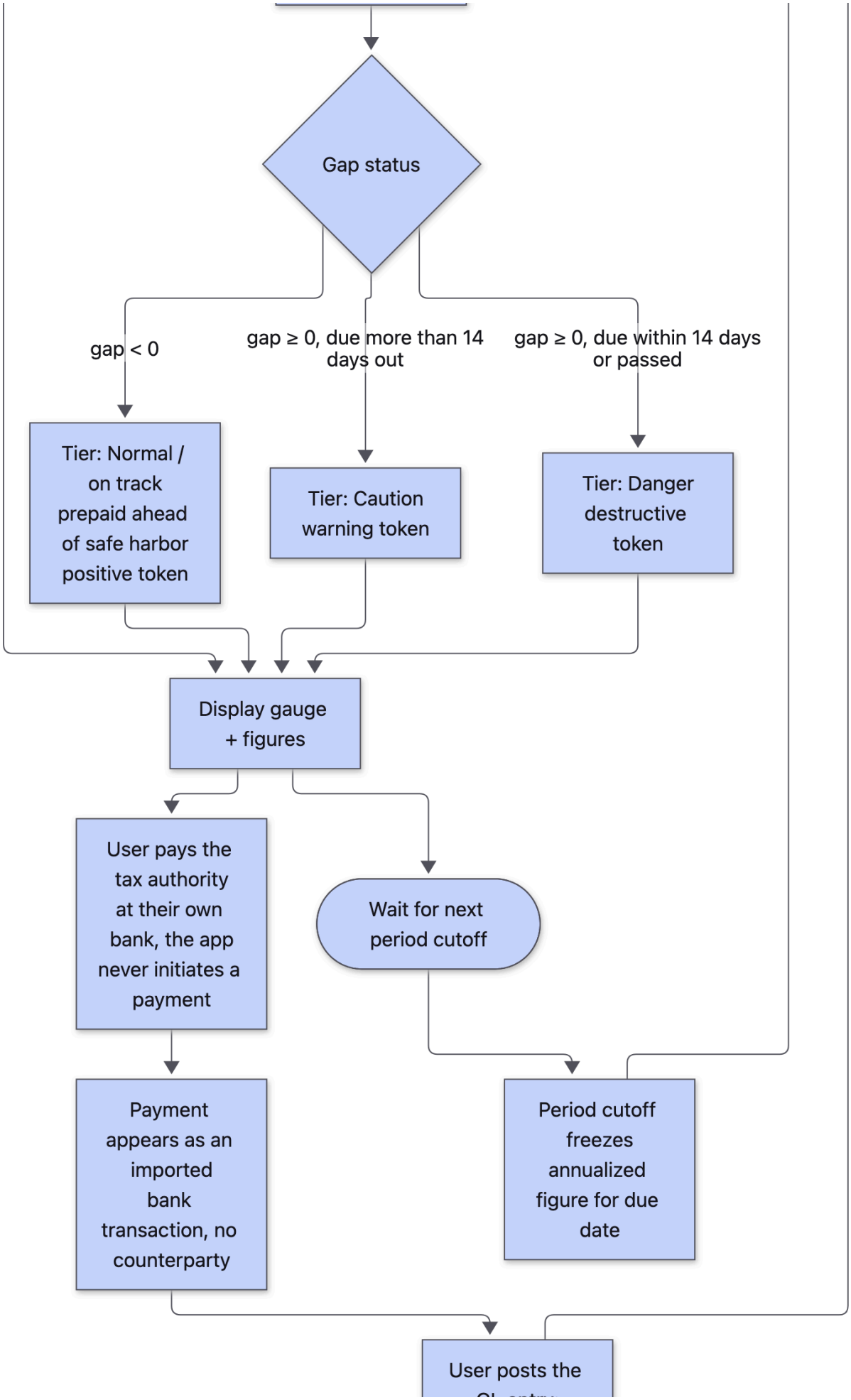
- **Mirror on Pending.** The same rule and window run on Pending: an imported transaction that matches a posted manual entry, whichever arrived first, shows a caution line on its row: "Looks like manual entry JE-000215 · Sep 11 · +$1,500.00". The line shows collapsed or expanded, and the Journal ID opens that entry's popover. The nearest date is named, and an entry already linked to an import is no longer a candidate. The review panel adds **Match to that entry** as its default action, and Approve & Post steps down to an outline button beside it so Enter cannot slip past it. **Match to that entry** links the import to the existing entry and posts nothing: the row leaves Pending, the entry stays as it is, and the Posted count holds. The entry's row gains a `matched import` status line and loses any `possible duplicate` badge, and its popover lists the import. Approve & Post past the caution keeps both. Dismiss possible duplicate on the entry clears the caution on the row it names.
- **Journal ID popover.** Clicking a Journal ID on Posted opens a `Popover` anchored to the ID, so an entry can be read without leaving the list. Header line: Journal ID with copy, transaction date, actor, "logged <timestamp>", status badge. Body: every leg, then the correction trail if any (the original and its reversal, since the entry's own legs are already above). Footer: **Open in General Ledger**, to the entry's row on the Journal entries card, plus **Dismiss possible duplicate** on a row that carries that badge. It is non-modal: an outside click or Esc dismisses it and focus returns to the ID. The View correction chevron on a Corrected row is a separate control: it expands the trail in place. The account register's Journal ID stays a link to the GL.
- **An unmatched transfer leg stays a staged draft** in Pending until its counterpart arrives (then it matches like any transfer), or until the user explicitly approves it into Cash in Transit as a one-sided in-flight item — a balanced entry, cash leg against the CIT clearing account, cleared when the counterpart posts. A draft still unmatched past the exceptions threshold moves to Exceptions.
- **Posting gate:** Σ debits = Σ credits; the accounting equation holds per entity; the entry date matches the bank date (or the Cash-in-Transit rule); every posted draft links to exactly one entry, and several drafts may share it (a group, a same-date matched transfer). Any failure keeps the item in Pending — no partial or silent post.
- A posted row's user-facing action is **"Correct entry"**, never "edit" or "reverse" — reversal-only underneath (PRD §5). Click-through shows the full trail: original lines shaded/struck, the reversal, and the new entry. Correct entry on any row of a multi-row entry corrects that one entry, and every row of it then carries the corrected entry's ID and the Corrected badge. See design-system-spec.md § Correcting a posted entry.

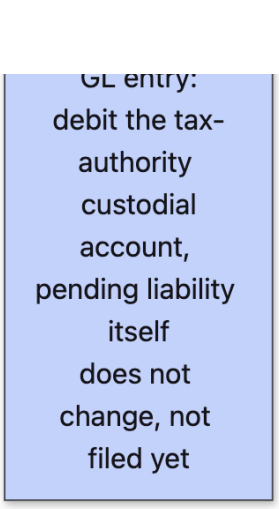
- Badges: `Draft (staged)` (`--secondary`, clock), `Posted` (`--outline`, check), `Corrected` (`--outline` + `--brand-2`-tinted dot, pencil). An auto-posted row carries `Posted` plus a `posted by rule — <name>` line and the same Correct entry action. `possible duplicate` (`--outline`, muted ink) sits under the status of a manual row posted past the duplicate warning; `matched import` is the status line of a manual row linked to its import.
- Actor attribution (user / auto-post rule / system job) is shown on every draft and posted row, never inferred (PRD §5).
- Import stats and link status (provider sync time, link health) show as a banner at the top of Transactions, not as a separate nav item.

5.4 Tax liability & safe-harbor

Story (PRD §4.5): *pending liability, required payment, gap, post your payment. Tax is a sidebar item. Source: flows/tax-liability-flow.md.*


Sidebar: Tax
Dashboard: Pending tax liability card
Tax view for the entity: jurisdiction tab selected
Running accrual annualizes income to date by tax treatment
Two safe-harbor bases computed: prior-year and current-year
Pending liability = annualized tax to date minus custodial prepaid balance
Required payment to date = lesser of the two safe-harbor bases
Gap = required payments to date minus custodial balance, are you current with the authority

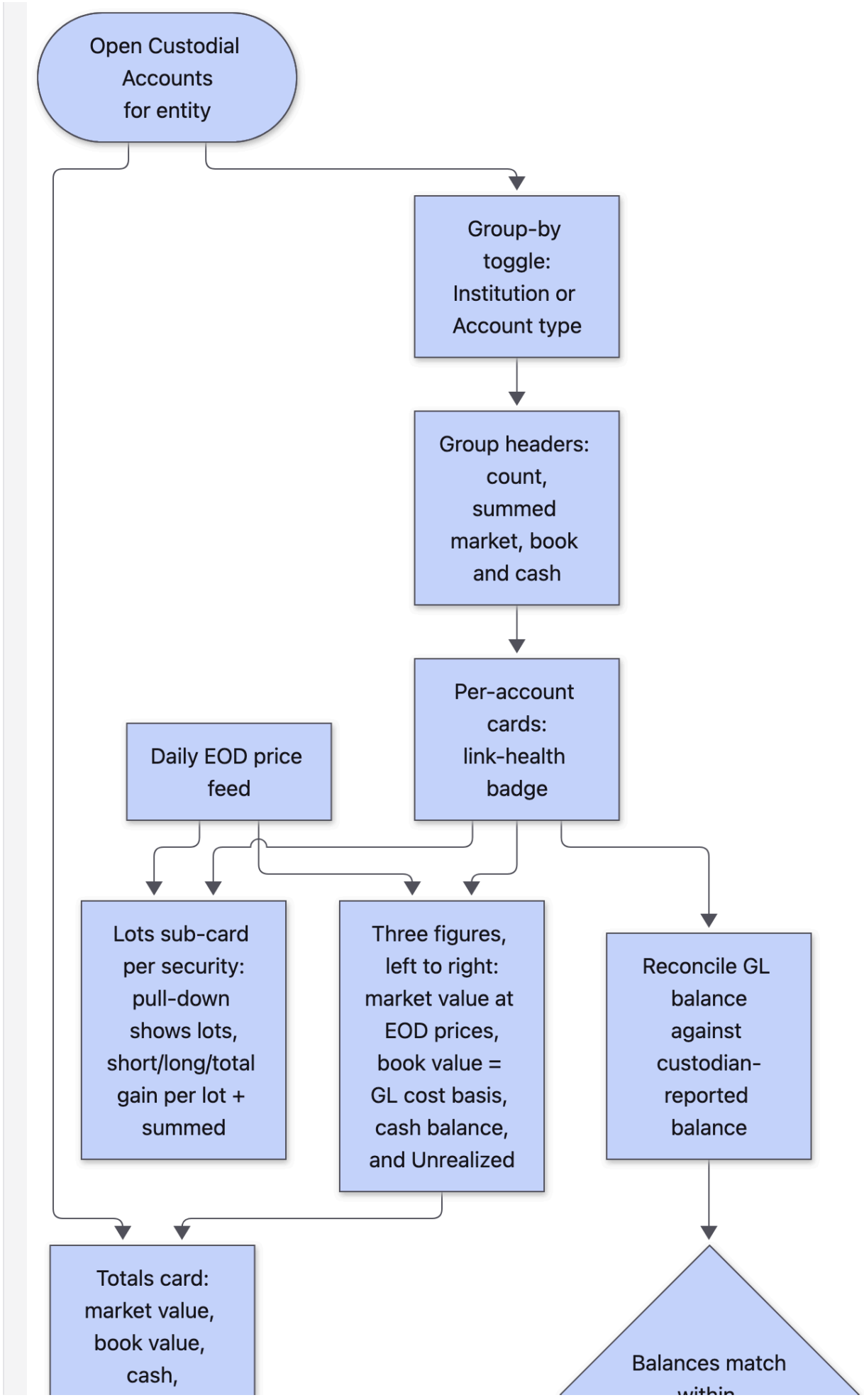

Gap status
gap < 0
gap ≥ 0, due more than 14 days out
gap ≥ 0, due within 14 days or passed
Tier: Normal / on track prepaid ahead of safe harbor positive token
Tier: Caution warning token
Tier: Danger destructive token
Display gauge + figures
User pays the tax authority at their own bank, the app never initiates a payment
Wait for next period cutoff
Payment appears as an imported bank transaction, no counterparty
Period cutoff freezes annualized figure for due date
User posts the

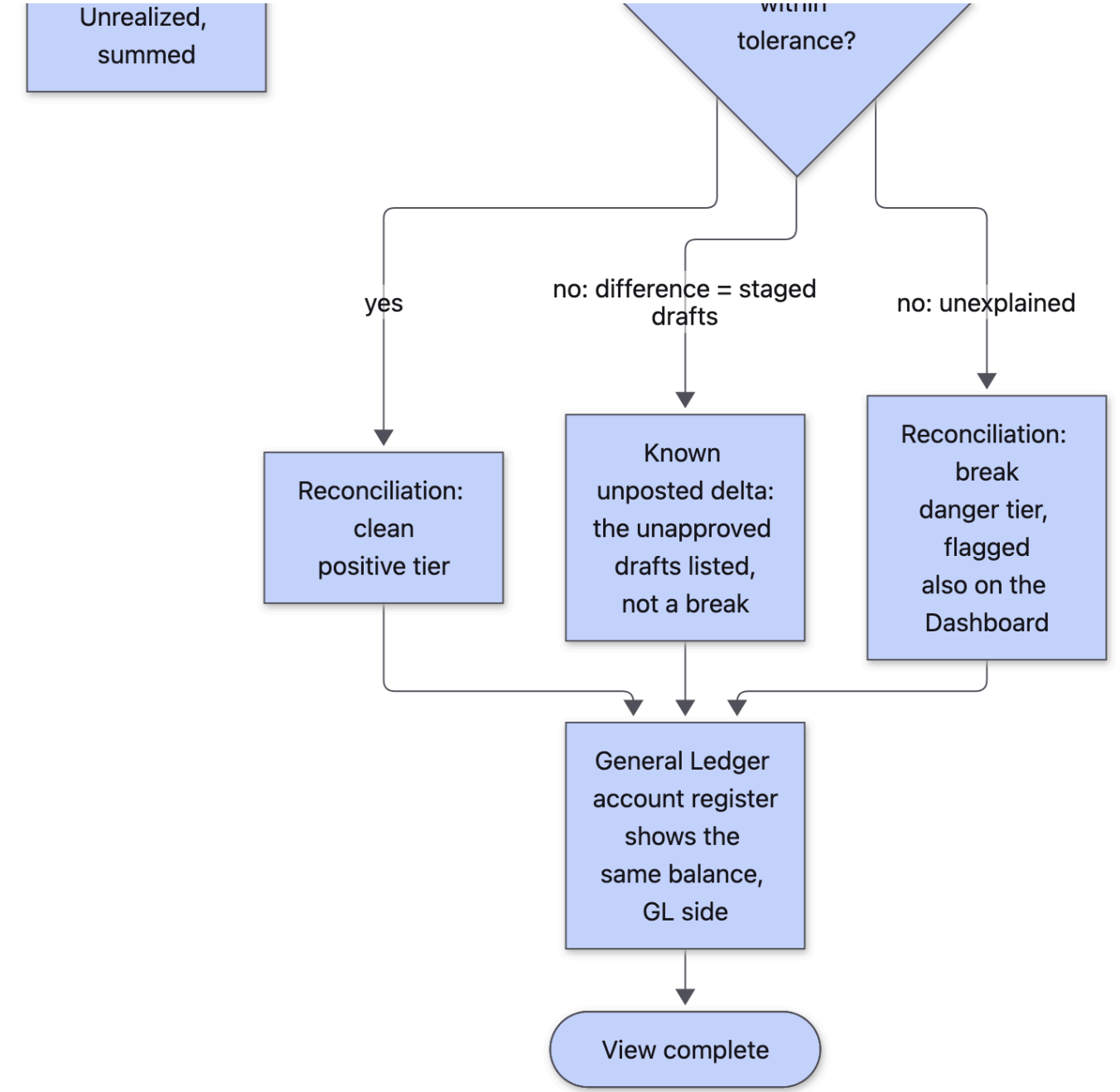


- **Reached from the sidebar or the Dashboard card;** both land on the same per-entity view with a jurisdiction selector.
- **Tiers:** on track when `gap < 0` (prepaid ahead of the safe-harbor requirement); caution when `gap ≥ 0` and the due date is more than 14 days out; danger when `gap ≥ 0` and the due date is within 14 days or has passed. They reuse the product-wide status vocabulary — design-system-spec.md § Status tiers.
- **Gap = 0 is not on track.** It means exactly at the safe-harbor line, with nothing in hand beyond the requirement, so it takes the caution or danger tier by due date.
- **Two figures,** both measured against the custodial account's balance but not against each other: **Pending liability = annualized tax to date – custodial prepaid balance** (what you would owe net of prepayments, if filed today); **Gap = required payments to date (safe harbor) – custodial balance** ("are you current with the authority"). Gap is never `required – pending`.
- **v1 never moves money.** There is no pay button: the user pays the tax authority at their own bank; the payment lands as an imported transaction with no counterparty; the user posts the GL entry, debiting the custodial tax-authority account (a prepaid-tax asset). Copy says "post your payment" or "record your payment", never "pay" or "transfer".
- Jurisdiction due dates, multipliers and cumulative percentages are data-driven (a lookup table), so the view renders for any number of jurisdictions per entity, not just federal (PRD §5).
- At filing, the frozen per-period figures export for Schedule AI / Form 2210 and the state equivalent; the flow's freeze step is the source of that export.

5.5 Custodial reconciliation & lots

Stories (PRD §4.6–4.7): *three figures per account, lots, and GL vs. custodial reconciliation. General Ledger and Custodial Accounts are two nav items, not a toggle. Source: flows/securities-lots-flow.md.*


Open Custodial Accounts for entity
Group-by toggle: Institution or Account type
Group headers: count, summed market, book and cash
Per-account cards: link-health badge
Daily EOD price feed
Lots sub-card per security: pull-down shows lots, short/long/total gain per lot + summed
Three figures, left to right: market value at EOD prices, book value = GL cost basis, cash balance, and Unrealized
Reconcile GL balance against custodian-reported balance
Totals card: market value, book value, cash,
Balances match

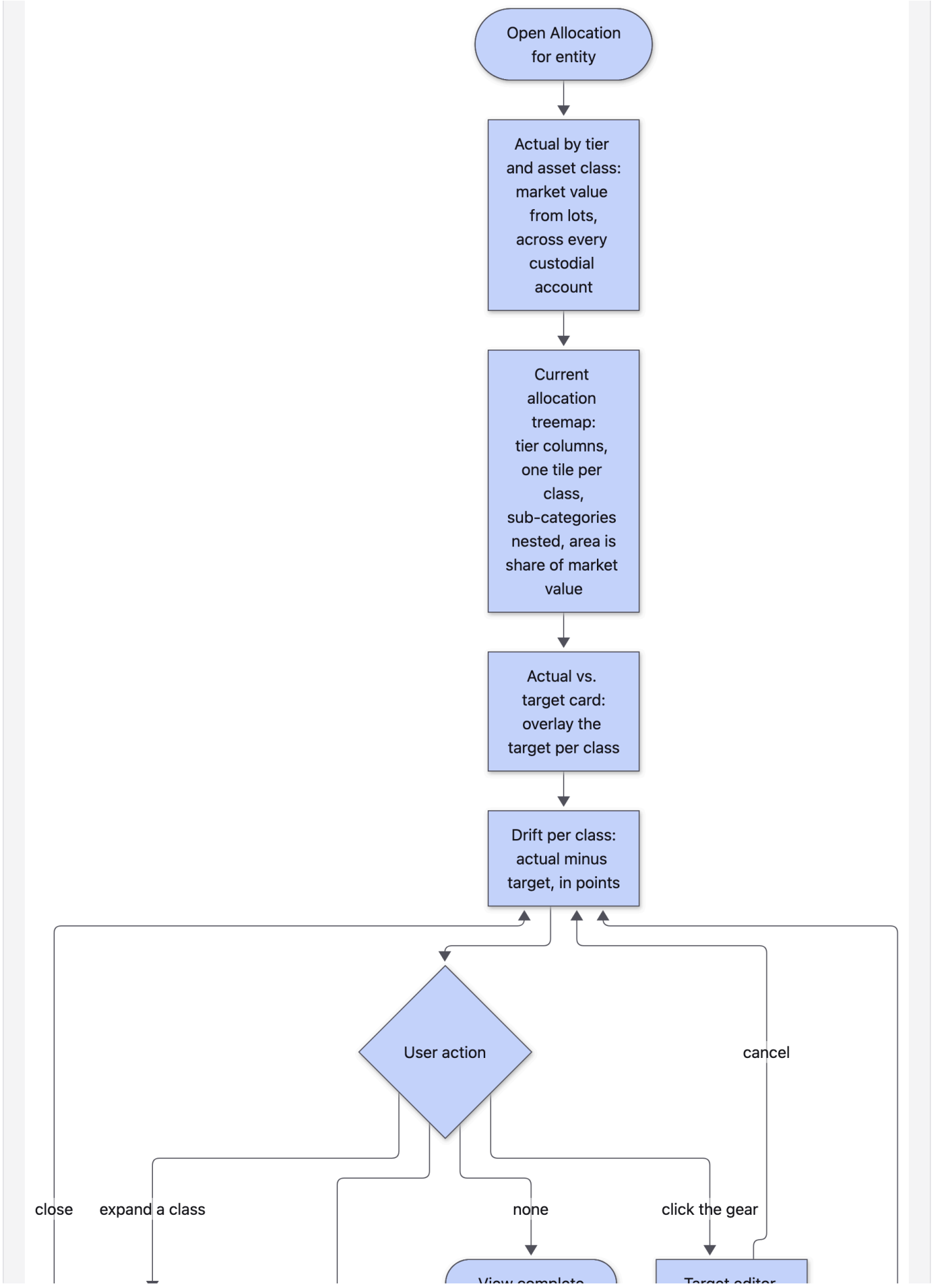


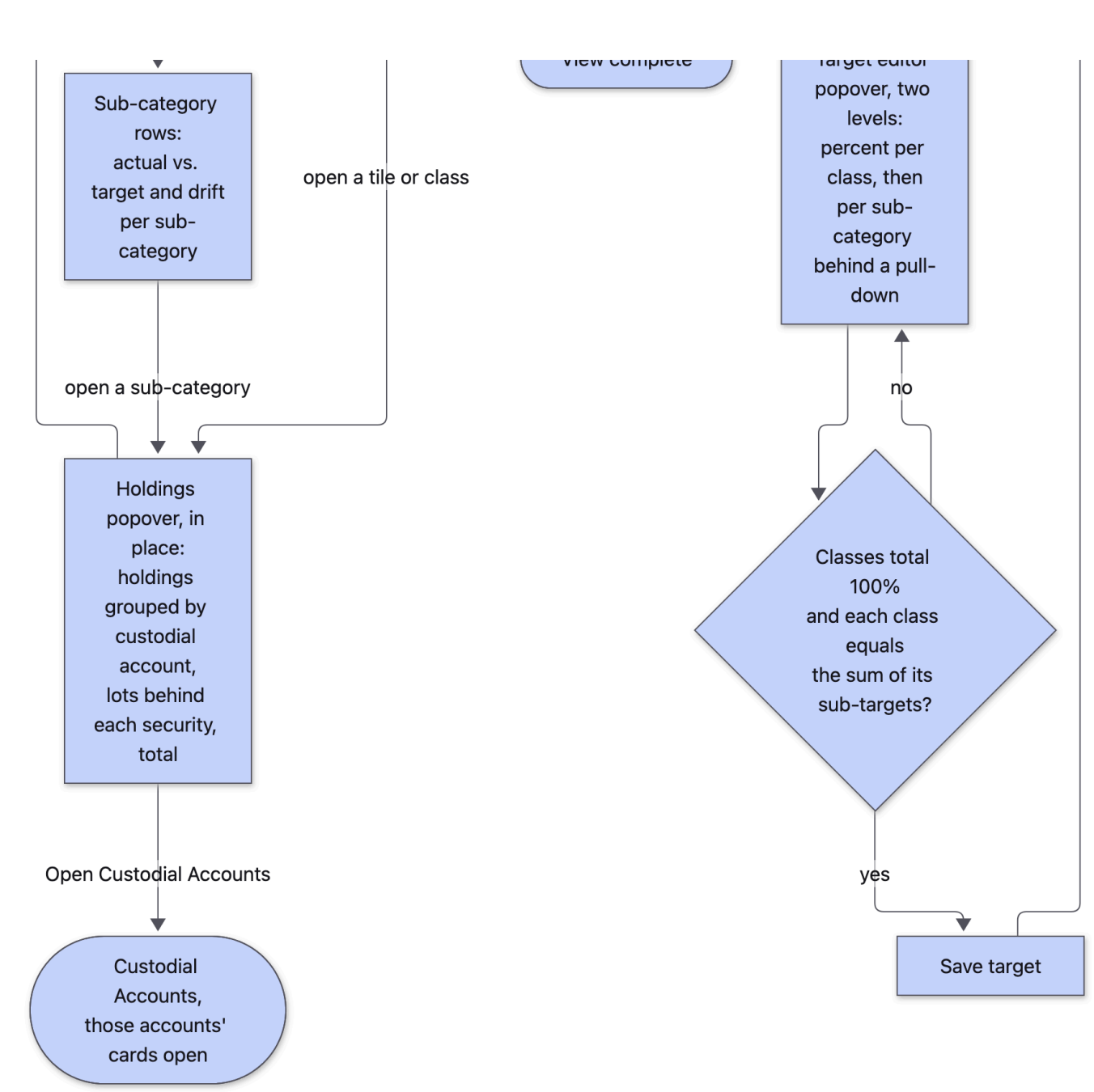
- **Three figures on every account, left to right:** market value (cash plus every position at the last EOD price); book value (the GL's cost basis, cash included — the GL's own balance for the mirrored account); cash balance; and Unrealized = market − book, signed and coloured per § Delta coloring. A cash-only account shows the same row with equal figures and a `$0.00` delta, so cards scan as columns. A totals card sums the four figures across accounts, with the price date. Cash sits last so the row reads as "market and book include cash" — the two totals first, the cash already inside them after.
- **Group by Institution or Account type.** A `Tabs` toggle under the totals card groups the account cards; Institution is the default, and the choice persists per user. A group header carries the account count and the summed market value, book value and cash. The totals card sums asset accounts whichever grouping is on.

- **A credit card is a liability.** Market value and book value both show what is owed, as a negative; cash and Unrealized show a dash. The negative nets into its group, and the totals card states it beside the asset total.
- **The card rests collapsed.** The header carries two badges in a fixed order, link health then reconciliation, because a reconciliation is only as current as the link behind it. A chevron left of the title opens the card's details: the custodian's figure against the GL and the link to any staged drafts, then, on an account that holds securities, the lots sub-card with its pull-down per security. Allocation's holdings popover links here and arrives with the cards of the accounts it lists open.
- An account's Unrealized figure equals its lots sub-card's total unrealized gain; the two must agree.
- **Two nav items:** General Ledger and Custodial Accounts are two separate nav items, not one screen behind a `Tabs` switch. Both read the same lot and journal data; the split is presentation, not a data fork.
- **Unapproved drafts are a known unposted delta, not a break.** When the GL and custodian balances differ by exactly the account's staged drafts (e.g. an unmatched transfer leg not yet approved into Cash in Transit), the card shows the delta and lists those drafts, linking to Pending on Transactions; no danger tier. Only a difference the drafts do not explain is a break.
- A reconciliation break uses the danger tier (`--destructive`) and surfaces on the Dashboard glance too, not only here.
- Allocation (actual vs. target by asset class) is its own flow, 5.6.

5.6 Allocation vs. target

Story (PRD §4.8): *the current allocation as a treemap, then actual vs. target by asset class and sub-category with drift; a two-level target editor behind a gear. Source: flows/allocation-flow.md.*


Open Allocation for entity
Actual by tier and asset class: market value from lots, across every custodial account
Current allocation treemap: tier columns, one tile per class, sub-categories nested, area is share of market value
Actual vs. target card: overlay the target per class
Drift per class: actual minus target, in points
User action
close
expand a class
none
click the gear
cancel



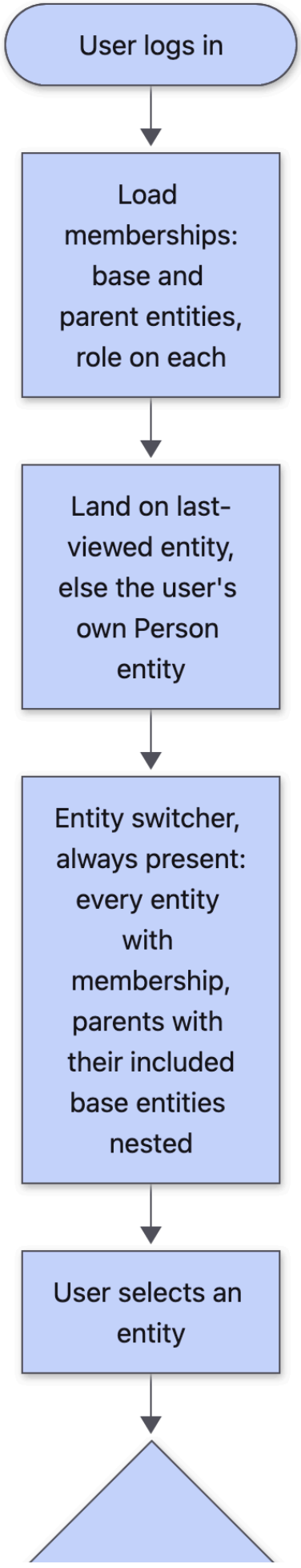
- **Same data, different aggregation:** actual allocation is the lots' market value summed by asset class across every custodial account for the entity — not a new data source.
- **Drift is information, not a warning.** Over or under target shows as a neutral `outline` badge ("+2 pt over", "−2 pt under"); on target shows the positive soft badge. No caution or danger hue.
- **Tier, class, sub-category.** Asset classes sit under two tiers: Current (Cash & Cash Equivalents) and Non-Current (Fixed Income Securities, Marketable Securities, Alternative Investment Assets). A tier is a label with its share, never a colour: captions over the treemap's two columns, group headers in the bars and the target editor.
- **Treemap first, table second.** The page opens on the current allocation as a nested treemap: one tile per asset class in its fixed categorical slot, its sub-categories inside it in the same slot.

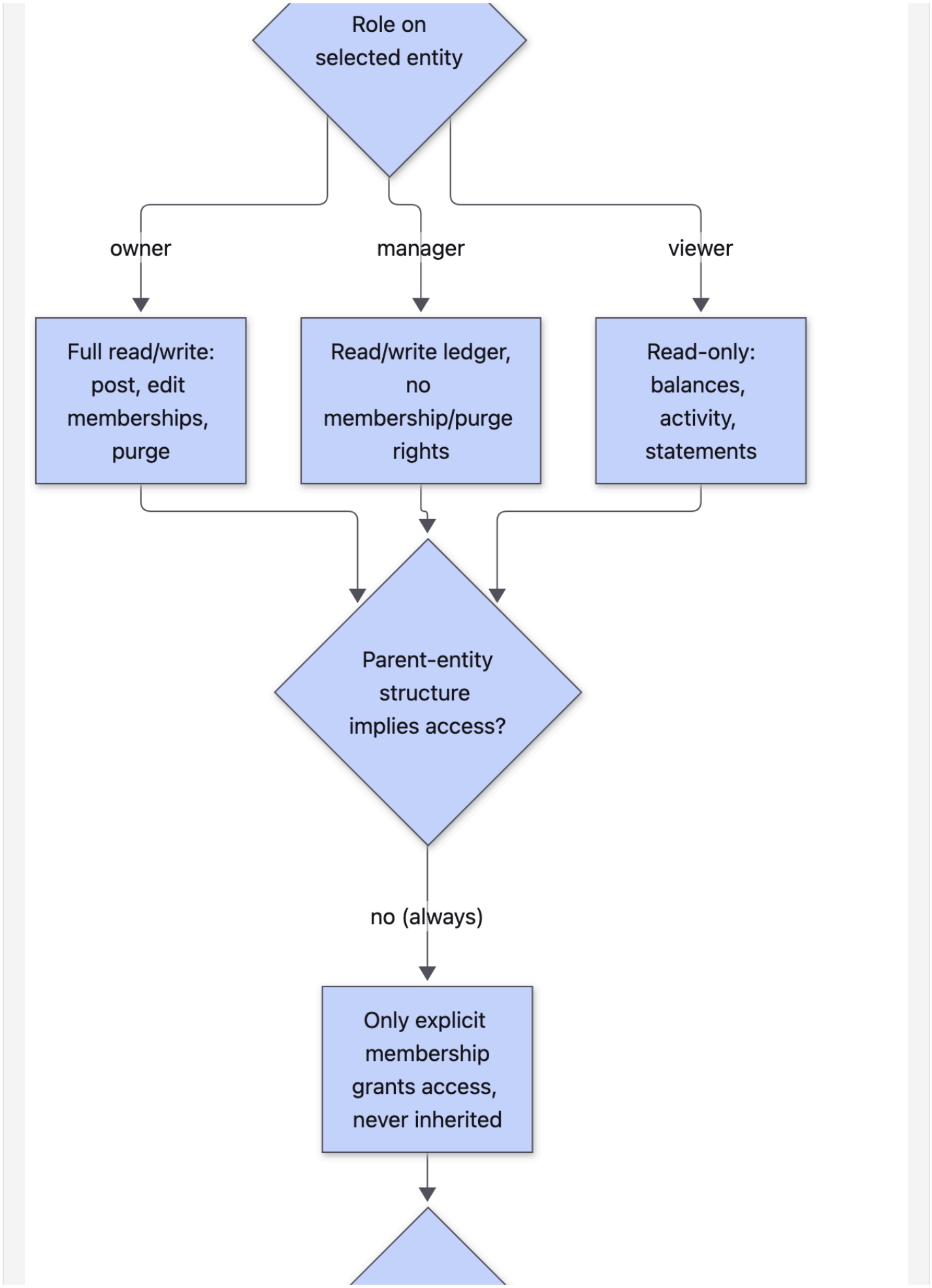
The Actual vs. target card below is its table view and carries every percent.

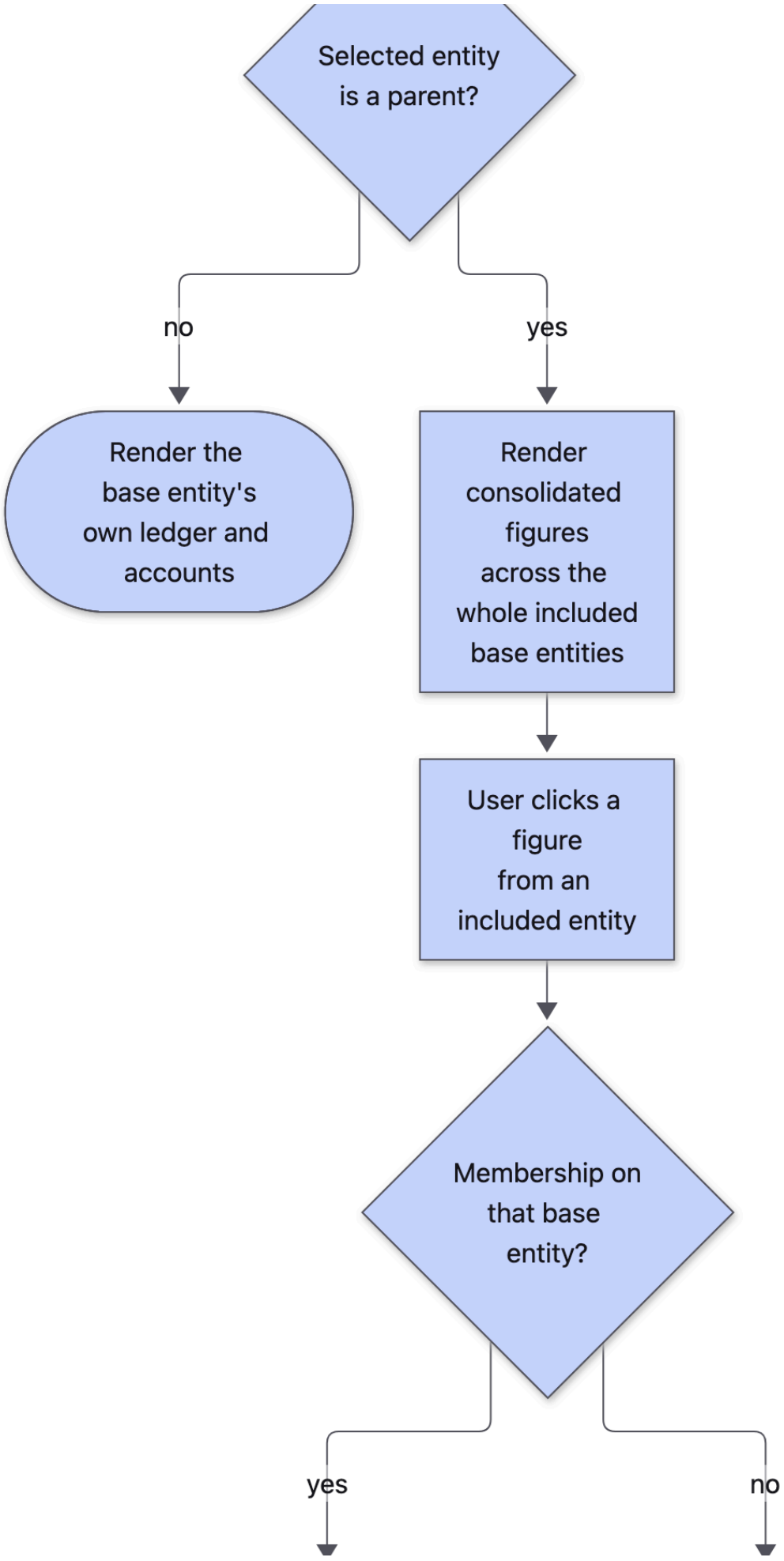
- **Drill opens in place.** Each treemap tile and each asset-class or sub-category name opens a read-only holdings popover on Allocation, because a class spans accounts and no single Custodial card can show it. Header: name, tier path, actual percent and dollars, target, drift badge. Body: the holdings behind that figure, grouped by custodial account with the account's share, each security with the Custodial card's lots pull-down; a cash sub-category lists cash balances per account. Footer: the aggregate total and one link, Open Custodial Accounts, which arrives with those accounts' cards open. Esc or a click outside closes it.
- **Sub-categories behind a pull-down.** Every class row, in the Actual vs. target card and in the target editor, has a chevron that reveals its sub-categories (the lots sub-card's disclosure pattern). Seeded defaults: Cash & Cash Equivalents → Cash, Money market; Fixed Income Securities → Treasuries, Corporate, Municipal; Marketable Securities → US Large cap, US Mid/Small cap, Intl Developed, Intl Emerging; Alternative Investment Assets → Digital Assets. In the Actual vs. target card, each sub-row shows its own actual / target and drift badge.
- **The gear opens the target editor.** A gear icon button at the top right of the Actual vs. target card opens the editor in a popover anchored to it, non-modal so the bars stay readable while targets change. Cancel or Esc closes it and discards the edits; a click outside closes it only when nothing has been edited. Focus moves to the first input on open and returns to the gear on close.
- **Target editor, two levels:** sub-targets are whole-portfolio percent and must sum to their class target; the class row stays editable and carries a reconcile badge (positive soft "Sub-targets sum to N%", or neutral outline "N pt unassigned"). Save is enabled only when the classes total 100% and every class reconciles. Viewers see the treemap and the bars but no gear (write controls are omitted, not greyed).
- **No nominal / inflation-adjusted toggle here.** Allocation is a contemporaneous snapshot of today's holdings, so a CPI-U restatement adds no information; the toggle (PRD §4.10) belongs to the time-series views (Dashboard net-worth chart, Reports).

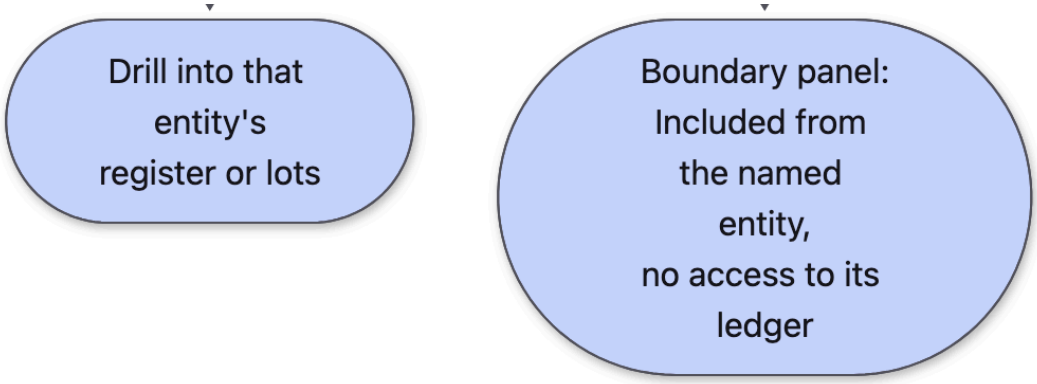
5.7 Entity switch, roll-up & membership access

Story (PRD §4.9): *parents roll up whole base entities; owner / manager / viewer per entity; the drill-through boundary. Source: flows/entity-membership-flow.md.*


User logs in
Load memberships: base and parent entities, role on each
Land on last-viewed entity, else the user's own Person entity
Entity switcher, always present: every entity with membership, parents with their included base entities nested
User selects an entity


Role on selected entity
owner
manager
viewer
Full read/write: post, edit memberships, purge
Read/write ledger, no membership/purge rights
Read-only: balances, activity, statements
Parent-entity structure implies access?
no (always)
Only explicit membership grants access, never inherited

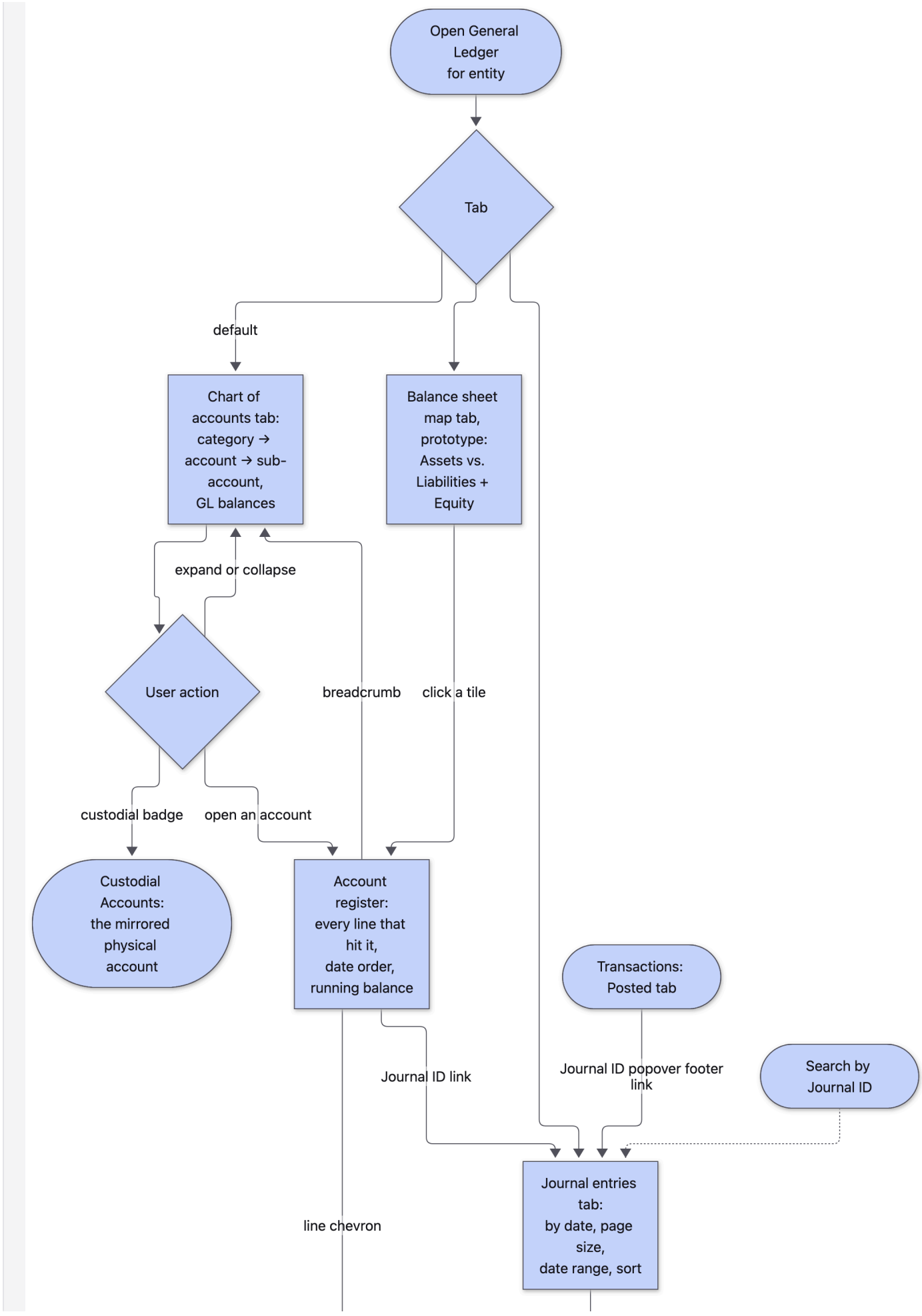

Selected entity is a parent?
no
yes
Render the base entity's own ledger and accounts
Render consolidated figures across the whole included base entities
User clicks a figure from an included entity
Membership on that base entity?
yes
no

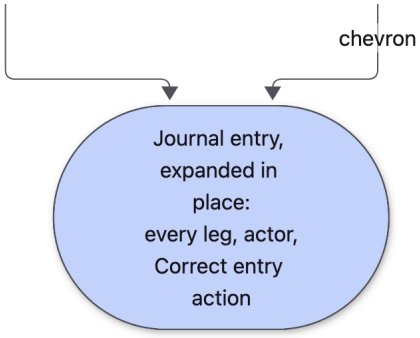


- **Base and parent entities.** Accounts (custodial and GL) live only in base entities: Person, Trust, Business. A parent (Household; Holding) rolls up **whole** base entities, never individual accounts. A base entity may belong to more than one parent.
- **The switcher is always present,** because every user owns at least their own Person entity. It lists every entity the user holds membership on, base and parent, with each parent's included base entities nested under it; a base entity in two parents is listed under both.
- **Access is explicit membership only** (PRD §5, hard requirement): a parent's membership never grants access to an included base entity, or vice versa.
- **The drill-through boundary.** A parent viewer sees the parent's consolidated figures. Every figure stays clickable; a drill into an included base entity the user has no membership on opens a boundary panel — "Included from <entity>. You don't have access to that entity's ledger." — instead of the register, never a disabled number or a silent dead end.
- Role badges (`Owner` / `Manager` / `Viewer`) use decreasing visual weight — `default` / `secondary` / `outline` (design-system-spec.md § Entity, account, and category badges).
- A viewer never sees a write affordance rendered-then-disabled: write controls are omitted entirely for a role that cannot use them.
- **Roll-up map:** Entities opens with a Pack of every entity the user holds membership on — parents as circles holding their base entities, each circle stroked in its entity type's categorical slot, a shared base entity drawn once in each parent with an "also in" marker; each circle switches entity. The membership list below it is the accessible fallback. See design-system-spec.md § Entities & membership.

5.8 General Ledger: tree, register, entry

Story (PRD §4.7): *three tabs; chart-of-accounts tree → account register → journal entry, with a custodial cross-reference. Source: flows/general-ledger-flow.md.*


Open General Ledger for entity
Tab
default
Chart of accounts tab: category → account → sub-account, GL balances
Balance sheet map tab, prototype: Assets vs. Liabilities + Equity
expand or collapse
User action
breadcrumb
click a tile
custodial badge
open an account
Custodial Accounts: the mirrored physical account
Account register: every line that hit it, date order, running balance
Transactions: Posted tab
Journal ID link
Journal ID popover footer link
Search by Journal ID
line chevron
Journal entries tab: by date, page size, date range, sort

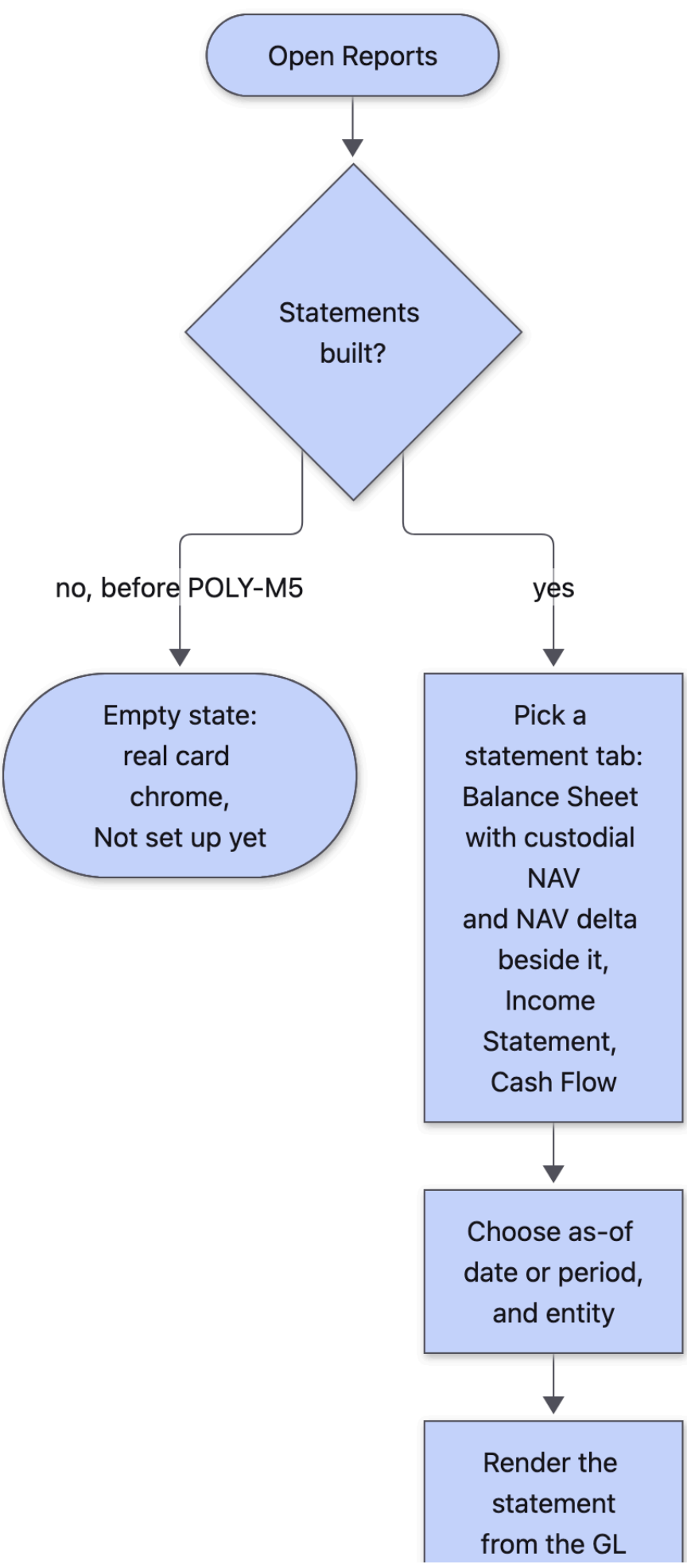


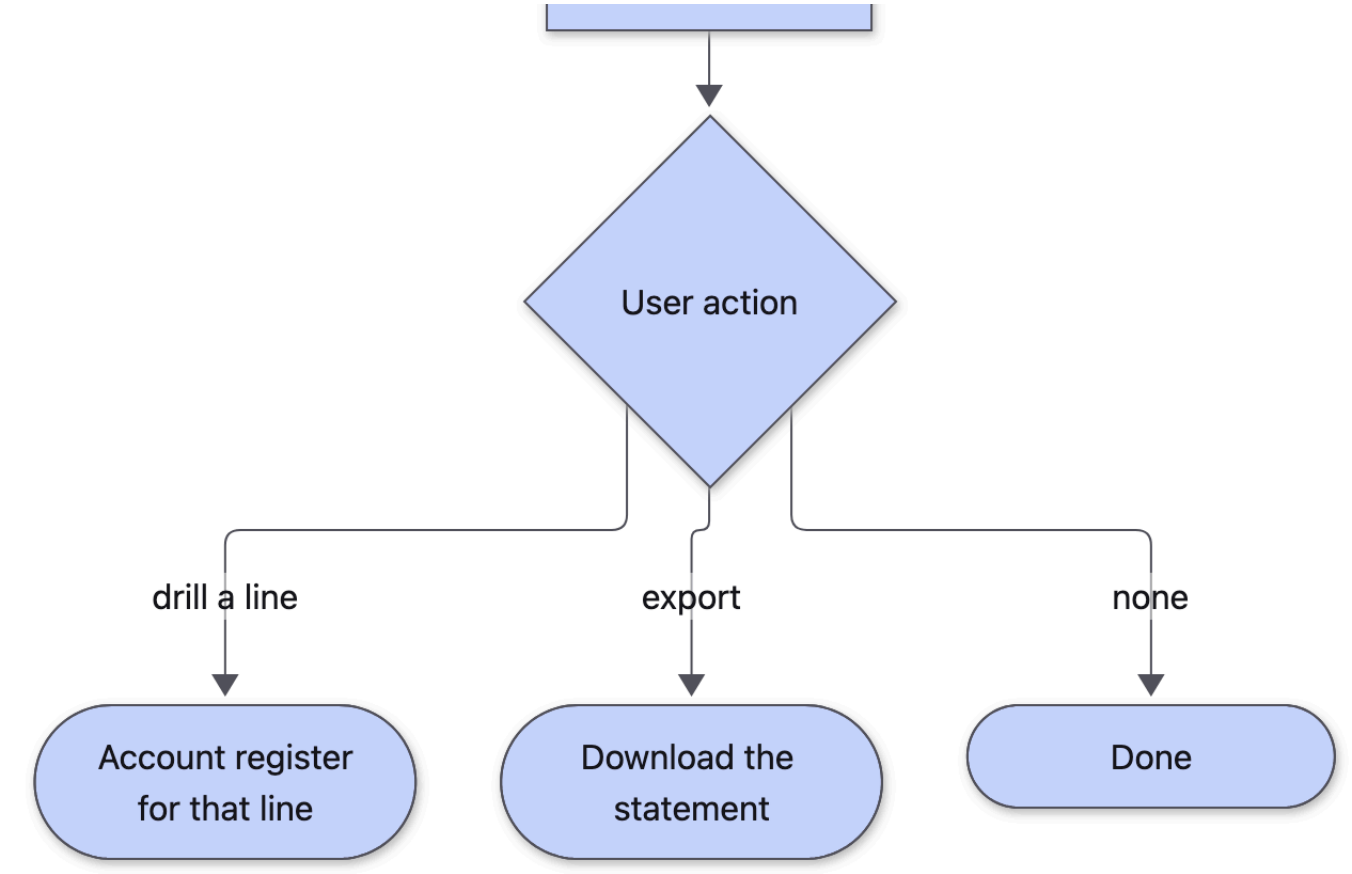
- **Three tabs, one section at a time:** Chart of accounts (the default), Balance sheet map and Journal entries, in one `Tabs` row at the top of the content. Each keeps its card header. The URL hash carries the tab, so Back from a register returns to the tab it left; a Journal ID link selects Journal entries; the register's breadcrumb returns to the tree.
- **The tree is the GL's own view:** every balance is the ledger's figure. The custodial badge (`landmark` icon + institution) is a cross-reference to Custodial Accounts, never a second data source; accounts with no physical counterpart (Owner's Equity, Cash in Transit) carry none.
- **The register is account-scoped and running-balance-shaped;** Transactions is transaction-scoped: every posted transaction is a row and carries its entry's Journal ID; an entry built from several transactions appears as several rows with one ID; the GL Journal entries card is the entry-grained view. Different questions, deliberately two screens.
- **A register line expands to its full journal entry** in the same block as the Journal entries card: every leg, not just the one that hit this account, plus the correction trail where one exists, with actor attribution and the Correct entry action (flow 5.3). Its Journal ID links to the GL card.
- **Balance sheet map (prototype):** a nested treemap on its own tab, Assets | Liabilities + Equity as equal halves, each element in its fixed categorical slot; each tile opens its register. Allocation's current-allocation map is the second instance of the pattern. Recommendation and limits: design-system-spec.md § General Ledger → Treemap prototype.
- **A leg always names its account by full path,** root to leaf, in the expanded entry on the register and on the Journal entries card: Assets › Cash › Primary Checking. The tree and the treemap show the leaf alone.
- **Journal entries card:** on its own tab, every posted entry header for the entity, newest first, with page size, date range (default This month, by ruling) and sort. A chevron opens the legs in the same block as Transactions' Posted tab. It lists a correction's three entries as three rows; the Posted tab rolls them into one.
- **Journal ID:** a short per-entity display number, `JE-000214`, assigned at posting and distinct from the UUID key: gapless, serialized per entity, never reused. It leads the Journal entries card, the Posted tab and the register, copied on hover, so any entry can be cited exactly.

- **Journal ID links:** on the register the ID links to the entry's row on the Journal entries card (`general-ledger.html#je-000212`), and on the Posted tab the ID opens a popover whose Open in General Ledger link goes to the same row (flow 5.3). Arriving selects the tab, highlights the row and opens its legs, widening the date range if needed. An exact ID in search resolves to the same row within the active entity; no search surface is drawn yet (design-system-spec.md § Header).
- **Reversed entries stay listed, muted:** no filter hides them. A Reversed original and its reversal render in muted ink, the original's description struck through; the Journal ID keeps full ink because it is the reference.

5.9 Reports

Stories (PRD §4.6–4.7): *pick a statement, scope it, drill, export. An empty state until POLY-M5 builds the statements. Source: flows/reports-flow.md.*

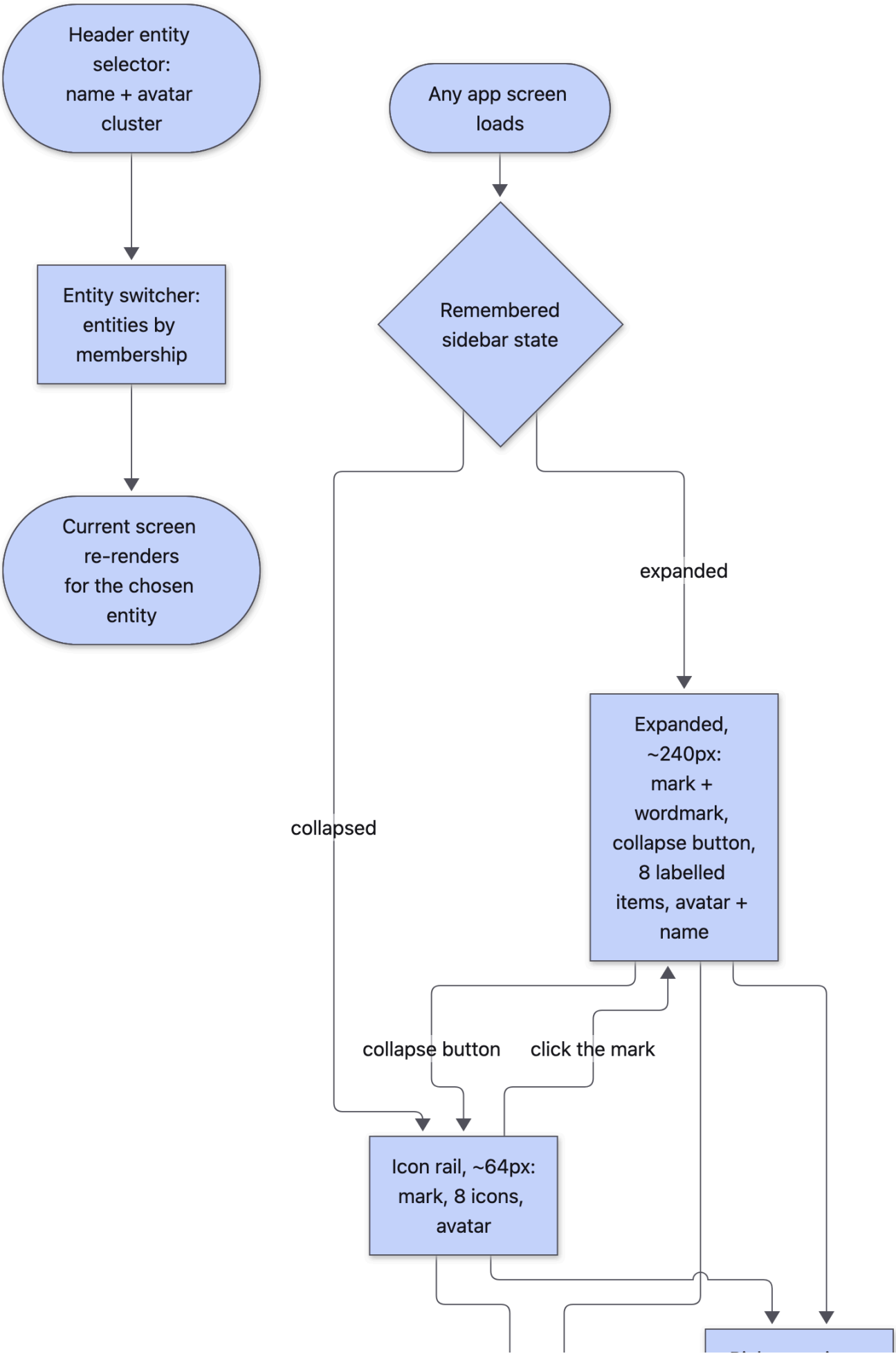

Open Reports
Statements built?
no, before POLY-M5
yes
Empty state: real card chrome, Not set up yet
Pick a statement tab: Balance Sheet with custodial NAV and NAV delta beside it, Income Statement, Cash Flow
Choose as-of date or period, and entity
Render the statement from the GL

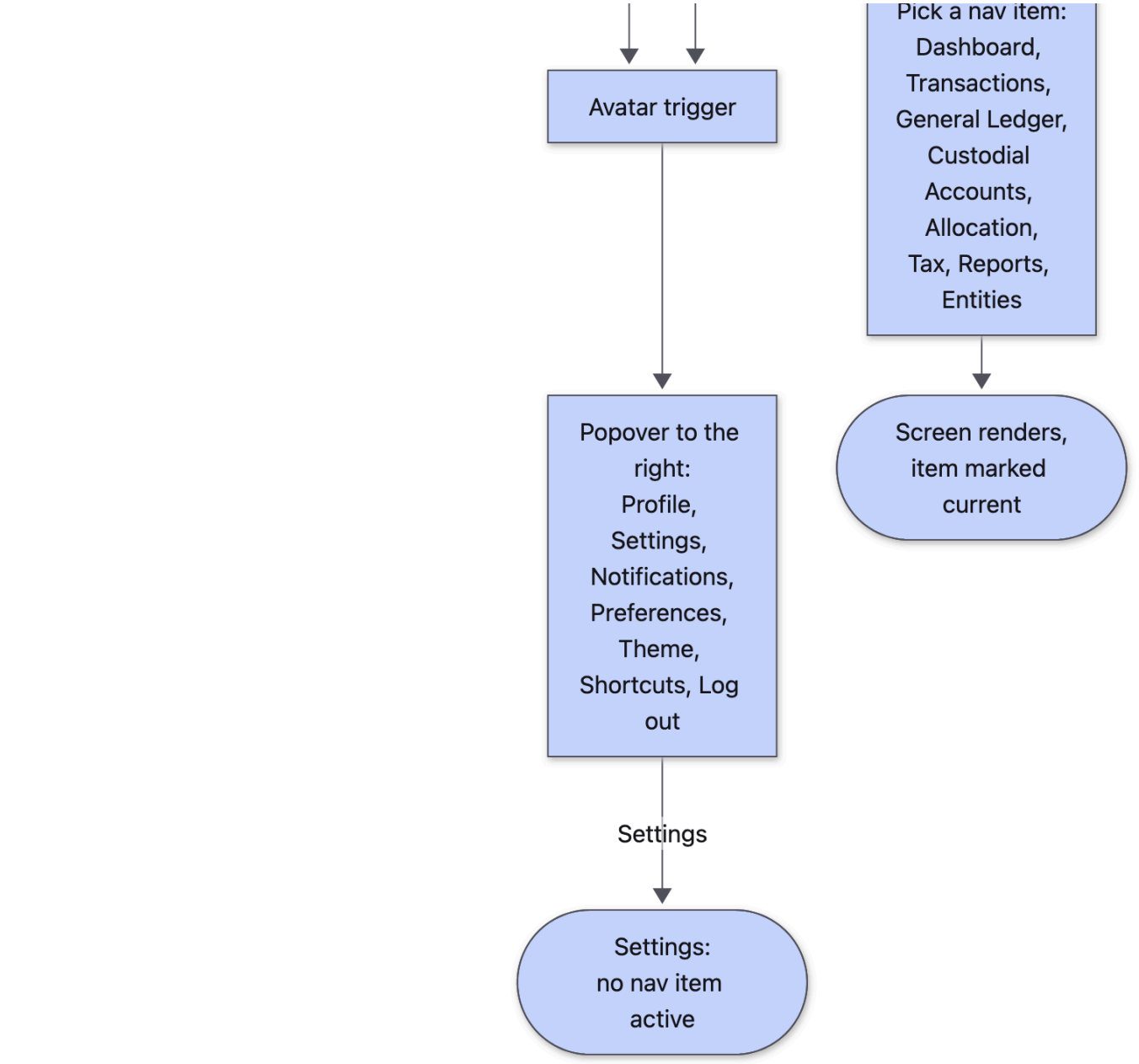


- **A nav item from day one.** Until POLY-M5 ships statements, every tab shows the Dashboard's empty-state pattern: real card chrome and plain "Not set up yet" copy that never names a milestone or version. The tab is never hidden or disabled.
- **Custodial NAV and NAV delta** (ARCH §2.4) sit beside the Balance Sheet, not as extra tabs.
- The as-of date (Balance Sheet) or period (Income Statement, Cash Flow) and the entity scope every figure; the entity comes from the header selector.
- Every statement line drills to the account register behind it (flow 5.8). Export formats are not yet specified.
- `reports.html` shows dimmed Income Statement and Cash Flow previews for review only; they do not ship.

5.10 Sidebar & app shell

Story: *the shell itself: eight flat items, icon-rail collapse, entity selector, avatar popover, keyboard. Source: flows/sidebar-flow.md.*


Header entity selector: name + avatar cluster
Entity switcher: entities by membership
Current screen re-renders for the chosen entity
Any app screen loads
Remembered sidebar state
expanded
collapsed
Expanded, ~240px: mark + wordmark, collapse button, 8 labelled items, avatar + name
collapse button
click the mark
Icon rail, ~64px: mark, 8 icons, avatar



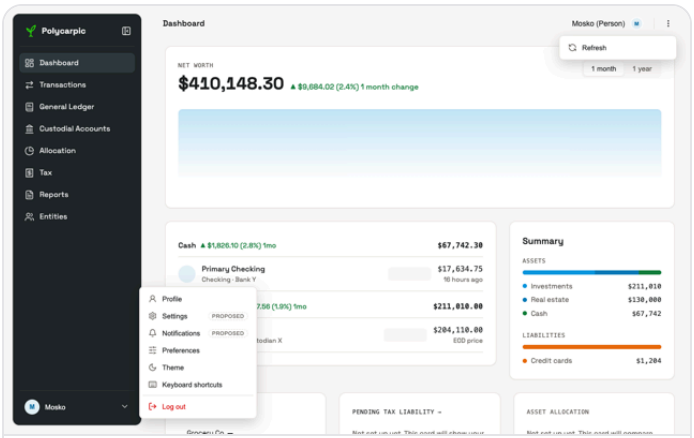
- **Eight flat items, no nesting** (design-system-spec.md § Sidebar information architecture). Flat is what makes the icon rail work.
- **Collapse to an icon rail.** The cue sits in the brand row: expanded shows a `panel-left-close` button right of the wordmark; collapsed shows the mark alone, which is itself the expand button and swaps to `panel-left-open` on hover or keyboard focus.
- **Remembered state:** the expanded or collapsed choice persists per user across screens and reloads (shadcn `Sidebar`'s stored `open` state).
- **Header entity selector:** the current entity's name and its member avatar cluster (max 3 + "+N"); it opens the entity switcher, and the current screen re-renders for the chosen entity (flow 5.7).
- **Avatar popover** opens to the right of the trigger, in both widths. Settings and Notifications are proposed entry points (`.proposed-tag`); Log out is last, `--destructive`-styled.

- **Keyboard:** Tab order is brand-row button → nav items → avatar trigger; the header follows in the main region. Enter or Space activates; Esc closes the popover and returns focus to its trigger. Every control shows the `--sidebar-ring` focus outline.

6. Screens

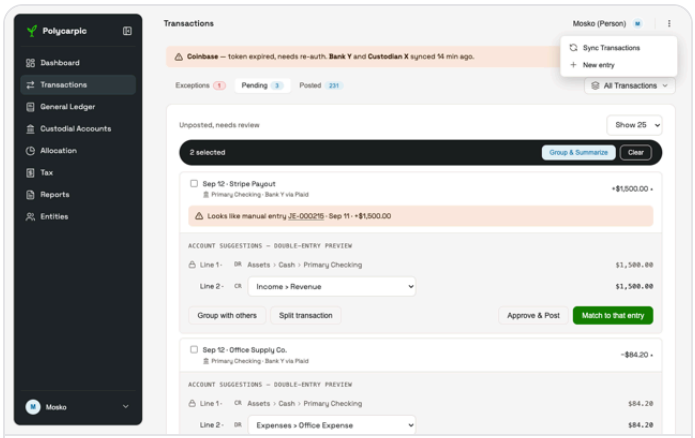
Every page, rendered with the real design system: `tokens.css` + `screen.css`, the sidebar-07 shell, and Lucide icons. These are the visual-regression reference and the head start for `frontend-lead`. There is one fidelity tier: every page is a styled screen. Each screen supports light and dark (`.dark` on `<html>`) and is checked at 960px and 1280px. Click a screen to open it; the sidebar nav links between them.

App screens



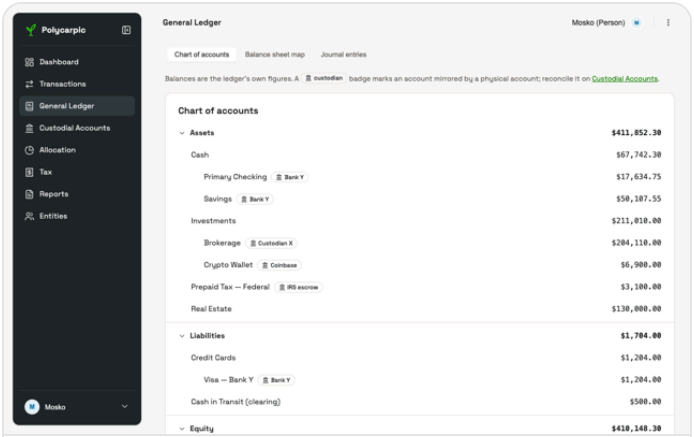
Dashboard

Per-entity financial health: net worth, accounts, summary, empty tax/allocation cards.



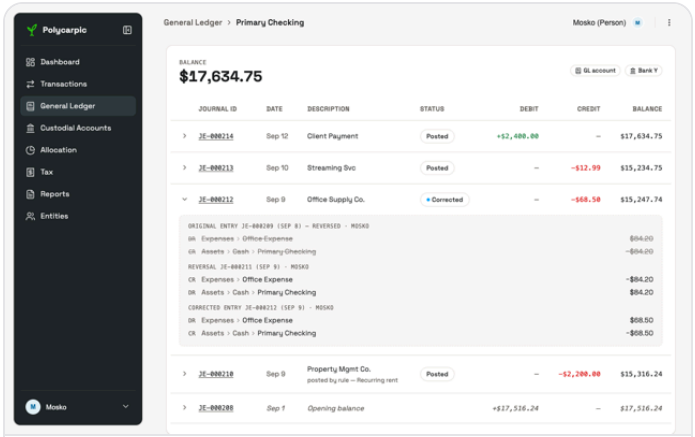
Transactions

Exceptions / Pending / Posted tabs with count badges, opening on the Plaid review-and-match inbox; split grid, bulk bar, Posted rows led by Journal ID, journal entry popover, journal-entry modal.



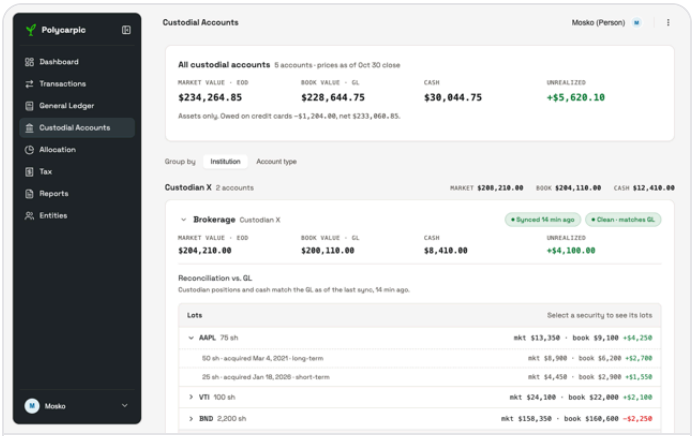
General Ledger

Three tabs: Chart of accounts (default) with custodial cross-reference badges, each account opening its register; Balance sheet map (prototype treemap); Journal entries with page size, date range and sort.



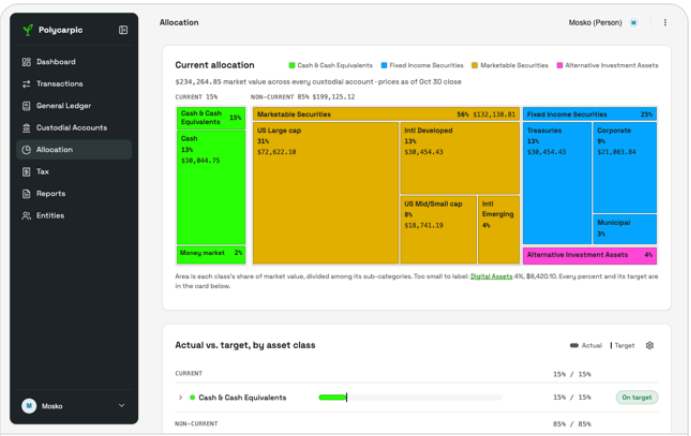
Account register

Drill-down from the GL: multi-crumb breadcrumb, Journal ID plus Dr/Cr/running-balance table, each line expanding to its full entry; Corrected and rule-posted rows.



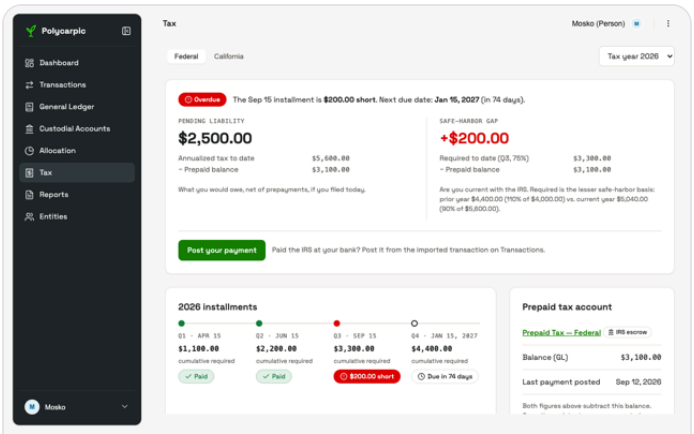
Custodial Accounts

Totals card, a group-by toggle (Institution or Account type), then per-account cards under group headers: link-health and reconciliation badges in the header, market value, book value and cash side by side, reconciliation details and a lots sub-card behind a chevron; the provider-item dialog shown open.



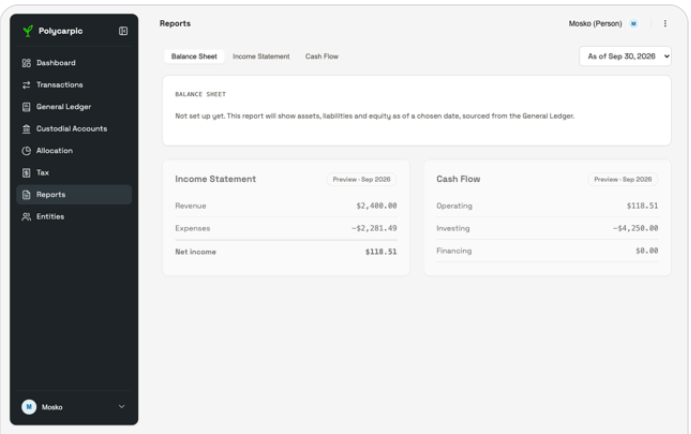
Allocation

Current allocation as a nested treemap on top; actual vs. target below, each class pulling down to its sub-categories and drilling to its lots; a tile or name opens its holdings across accounts in a popover, and a gear opens the two-level target editor in another (both shown open).



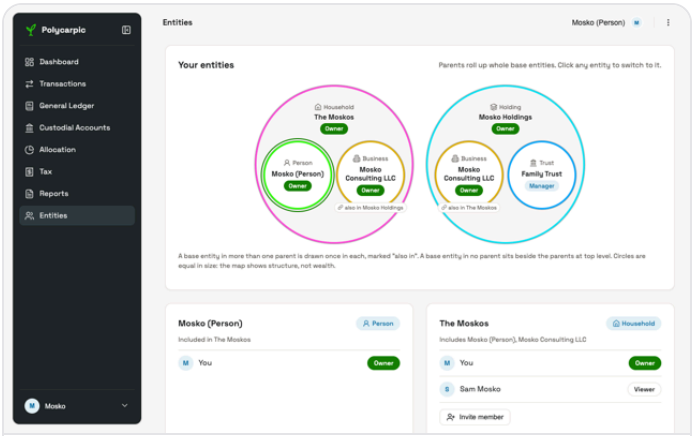
Tax

Per entity and jurisdiction: pending liability and safe-harbor gap side by side, status tier, installment timeline, prepaid-tax account; Post your payment.



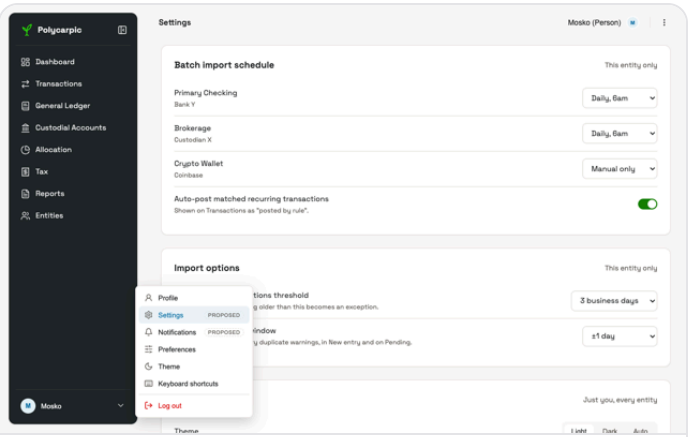
Reports

Balance Sheet empty state plus dimmed previews of the Income Statement and Cash Flow shapes.



Entities

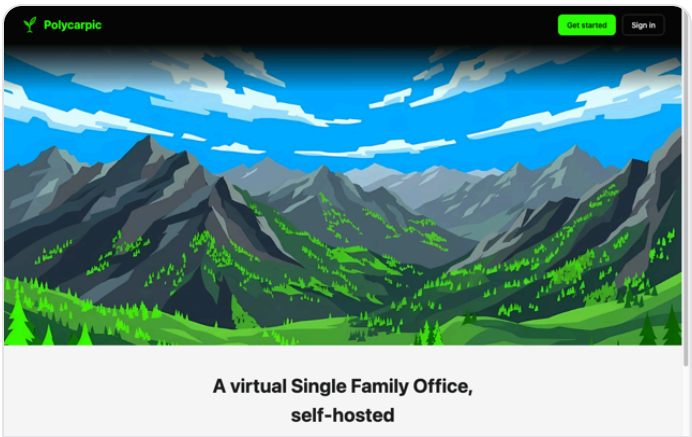
Roll-up map (Pack): parents holding whole base entities, a shared entity drawn in each parent; membership cards with Includes / Included in, role badges, invite and create below.



Settings

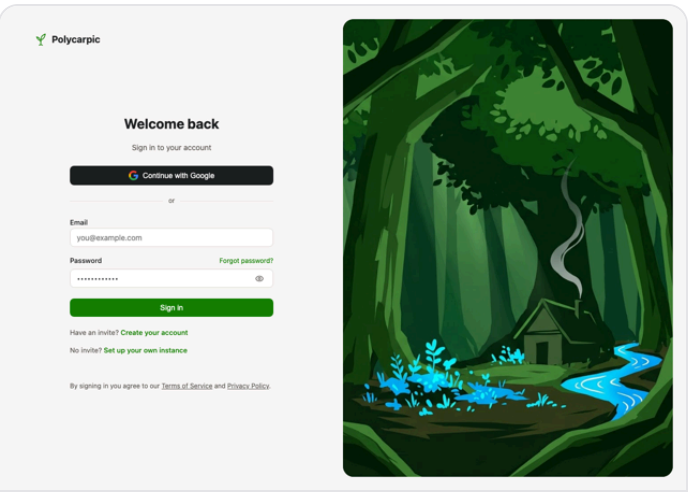
Per-entity import schedule and options, per-user preferences; avatar popover open with Settings current (proposed entry point).

Public & onboarding



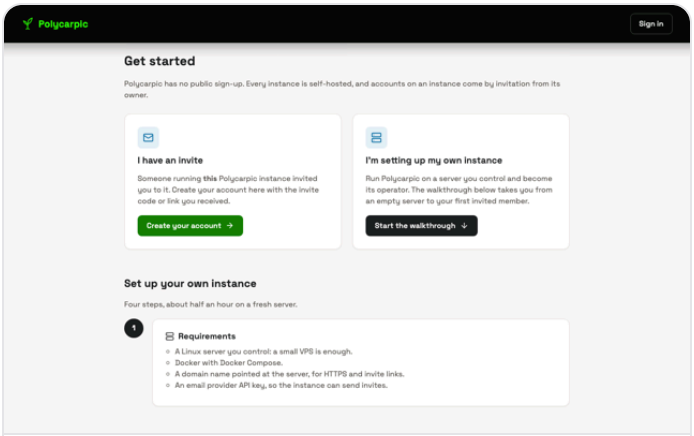
Home

Public pre-auth landing page: a fixed black top bar (neon lockup left, Get started + Sign in right) fading over the mountains illustration; headline, subhead and cards below.



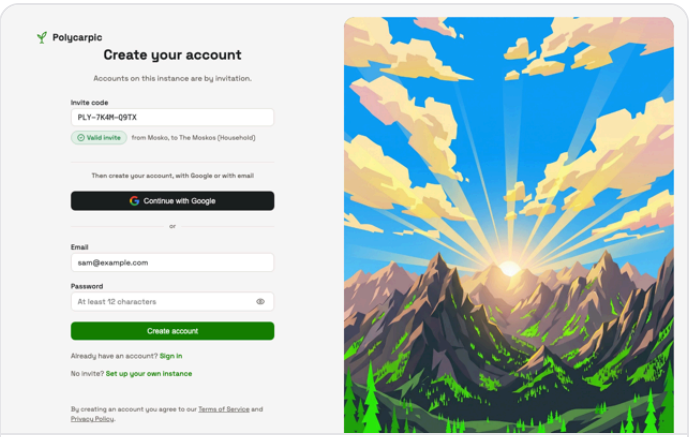
Login

Split screen: centered form pane, always-branded visual pane.



Get started

Where Home's Get started leads: a two-way fork (I have an invite / I'm setting up my own instance) with the self-host walkthrough below.

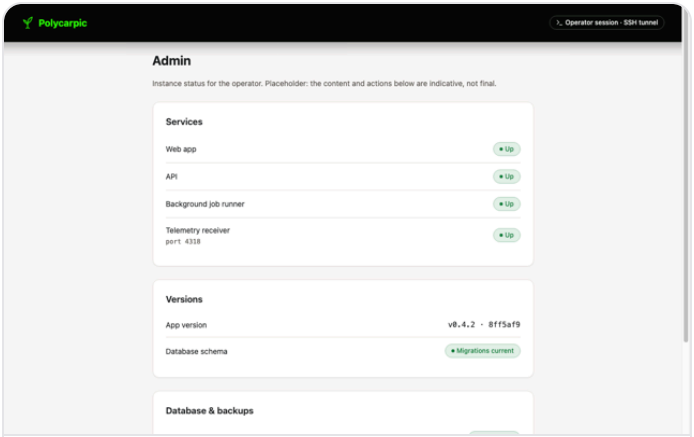


Sign up

Invite-gated account creation in Login's split layout, with its own sunrise pane; lands on the invited entity's dashboard.

6.1 Operator surfaces

Not reachable from any in-app nav and not part of the eight-item IA. Operator-only, reached by SSH tunnel to the host.



Admin

Placeholder, pending PRD/ARCH/SECURITY definition (POLY-81). Services, versions, database and backups, last sync; administrative actions shown disabled until defined.

7. Accessibility

Accessibility is a design responsibility, not a QA afterthought. Full table in design-system-spec.md § Accessibility.

Concern	Standard / approach
Color contrast	WCAG AA (4.5:1 text, 3:1 large/UI) — every token pair measured, both modes; `--brand` deliberately fails against light surfaces (1.36:1, by design — never used there, see Foundations); no CVD simulation run yet (open item, high priority given how central the brand hues are)
Color-alone signaling	Never — every gain/loss/status pairs color with a sign/icon/label
Keyboard	All interactive elements reachable & operable (shadcn default)
Screen readers	Semantic markup; labelled controls; delta sign readable from text content
Motion	Respect `prefers-reduced-motion` (policy not yet written)

8. Open Questions

Unresolved design decisions and assumptions that could be wrong. Batched to team-lead for relay to the Principal — not blocking.

- **Avatar cluster overflow:** the "+N" fold for 4th+ members is specified but not yet shown live in either reference screen (both currently have ≤2 members) — worth a quick visual check once a real multi-member entity exists.
- No CVD (color-vision-deficiency) simulation run on the palette yet — high priority given how central the brand hues are (recommended before lock, design-system-spec.md § Accessibility).
- Native iOS/macOS token export format (kickoff §2.10) — not yet decided, non-blocking for web.
- `prototype/` throwaway UX prototype (kickoff §2.9) not yet built — confirm ownership/timing with team-lead.

9. Appendix

- PRD user stories & functional requirements: docs/PRD/index.html §4–5
- Kickoff record, frontend/design system decisions: docs/project_kickoff.md §2.9

10.1 FAQs

Q: How do I add review comments to this design doc?

A: Hand-edit `docs/DESIGN/comments.md` with `## §<section-id>` blocks, or run `/serve-docs DESIGN` and use the inline `+ Comment` widget. Then run `/refine-doc DESIGN`. See process/WORKFLOW.md → Doc review loop.

Q: Why is the design system in `docs/DESIGN/` and not in the app's source?

A: It's the design *deliverable* — the contract. During Implement, `frontend-lead` imports or derives the app's real stylesheets from `tokens.css` / `screen.css`.

Q: Where did `variants.css` go?

A: It was generated infrastructure for the `cairn` tracker tool's own theme switcher, never part of the product — removed from this directory (POLY-78 relocates its generator target the same way). See design-system-spec.md § Provenance & shared tooling.

Generated by the `ux-designer` agent via `/generate-designdoc`. Edit by re-running the skill or editing in place.